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Empirical Profiling of Cloud-Based Healthcare Interoperability Gateways

Master Thesis

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Abstract

The digitalisation of community-based healthcare requires heterogeneous applications to exchange information through regional services whose behaviour must be assessed across complete transaction paths. In the context of SI.Ter (Sistema Informativo Territoriale), the Territorial Information System for the Healthcare Organisations of the Veneto Region, this thesis defines an experimental methodology for end-to-end performance profiling and differential overhead characterisation of cloud-based healthcare interoperability gateways. Documented regional workflows become reproducible workloads observed at the client boundary. The Request Dataset Generator (RDG) is a Python framework that creates synthetic Fast Healthcare Interoperability Resources (FHIR) R4 (Release 4) patients, composes ordered multi-step workflows, schedules configurable traffic, and concurrently executes independent flow occurrences. It supports the direct RVE (Regione Veneto) patient-registry transactions RVE-54 and RVE-55 and a gateway-mediated workflow in mock and live modes. Structured logs provide a traceable dataset for request- and flow-level latency, throughput, error-rate, concurrency, and scheduling metrics. Since internal gateway logs remain outside the authorised perimeter, the *Surrogate Observability Strategy* uses the `direct_vs_middleware` experiment to compare client-observed latency distributions for direct and mediated paths. Differences in means and selected percentiles estimate apparent path-level overhead, not internal gateway time; routing, security, transformation, network, and retry effects may also contribute. The empirical campaign comprises 74 completed runs. As a proof of concept, it distinguishes transport from application outcomes, request from flow populations, direct from mediated paths, and successful from failed processing. The contribution is a profiling framework and a reproducible surrogate-observability protocol for constrained healthcare integration environments.

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Acronyms and Abbreviations

APMS Application Profile Management System. 3

ATNA Audit Trail and Node Authentication. 7, 11

COT Centrale Operativa Territoriale. 1

FHIR Fast Healthcare Interoperability Resources. 4

FSE Fascicolo Sanitario Elettronico. x, 2, 7

HL7 Health Level Seven. 16

IAP Identity and Assertion Provider. 8

IHE Integrating the Healthcare Enterprise. 5

JSON JavaScript Object Notation. 4

JSONL JSON Lines. 16

JWT JSON Web Token. 6, 16

RDG Request Dataset Generator. 10

RVE Regione Veneto. 4, 15

SAML Security Assertion Markup Language. 6, 16

SI.Ter Sistema Informativo Territoriale. 3

XDS Cross-Enterprise Document Sharing. 3

Glossary

APMS A centralised system used by Azienda Zero to support user authentication and profile management for participating cloud-based applications. 3

ATNA An IHE integration profile that defines security measures for healthcare information systems, including node authentication, protected communications, and the generation and collection of audit records. 7, 11

COT A territorial coordination centre introduced in the Italian healthcare model to coordinate transitions among hospital, community, home-care, and other territorial services. 1

FHIR An HL7 standard for exchanging healthcare information electronically through modular resources, defined data models, and interoperable interfaces. 4

FSE The Italian *Fascicolo Sanitario Elettronico (FSE)*$_G$ infrastructure that makes health and social-care documents and data concerning an individual available to authorised actors. 2, 7

HL7 A family of standards for the exchange, integration, sharing, and retrieval of electronic health information. 16

IAP A regional security actor that authenticates a requesting application context and issues the identity assertion required by subsequent healthcare interoperability transactions. 8

IHE An international initiative that publishes integration profiles describing how established healthcare standards can be used together to support specific interoperability use cases. 5

JSON A text-based data-interchange format that represents structured data through objects, arrays, and primitive values. 4

JSONL A line-oriented data format in which each line is an independent JSON value, allowing structured event and execution records to be processed incrementally. 16

JWT A compact token format used to transmit claims between parties as a JSON object that can be digitally signed and, when required, encrypted. 6, 16

RDG The software framework developed in this thesis to generate synthetic healthcare data, compose interoperability workflows, schedule workloads, execute requests, and collect structured experimental observations. 10

RVE The acronym used as a prefix for interoperability transactions defined for the Veneto regional healthcare ecosystem. 4, 15

SAML An XML-based standard for exchanging authentication and authorisation assertions between identity providers and service providers. 6, 16

SI.Ter The Veneto regional initiative supporting territorial-care applications and their integration with local and regional healthcare systems. 3

XDS and XDS.b An IHE integration profile for registering, discovering, and retrieving clinical documents across healthcare organisations. XDS.b denotes the revised profile variant used by the interoperability architecture considered in this thesis. 3

Chapter 1

Problem Statement

1.1 Regulatory and Technical Context

1.1.1 Territorial Coordination

DM (Decreto Ministeriale) 77/2022 defines territorial care as a coordinated network of services rather than as a collection of isolated clinical encounters. Its organisational model includes the District, the Casa della Comunità, the COT (Centrale Operativa Territoriale), the 116117 operational centre, home-care services, community hospitals, and telemedicine-supported services[1]. The decree therefore assigns digital information exchange an operational role: coordination across care settings depends on information that remains available when responsibility for a patient moves between hospital departments, territorial services, general practitioners, and home care. The COT makes this dependency explicit because it coordinates transitions among care settings and connects actors that belong to different organisational and technical domains. Its work requires current patient-identification data, care-transition information, clinical documentation, and communication with local and regional services[2]. A protected discharge, for example, does not end when a hospital records the discharge decision: it continues through the transmission, assessment, assignment, and activation activities that make territorial care effective. The organisational process is consequently continuous even when the supporting applications are not.

[1] Ministero della Salute. *Regolamento recante la definizione di modelli e standard per lo sviluppo dell'assistenza territoriale nel Servizio sanitario nazionale.* Ministerial decree Decreto 23 maggio 2022, n. 77. Italian Official Gazette, General Series no. 144, 22 June 2022. Ministero della Salute, May 23, 2022, Allegato 1, pp. 12, 20–21.

[2] Ibid., Allegato 1, pp. 20–21.

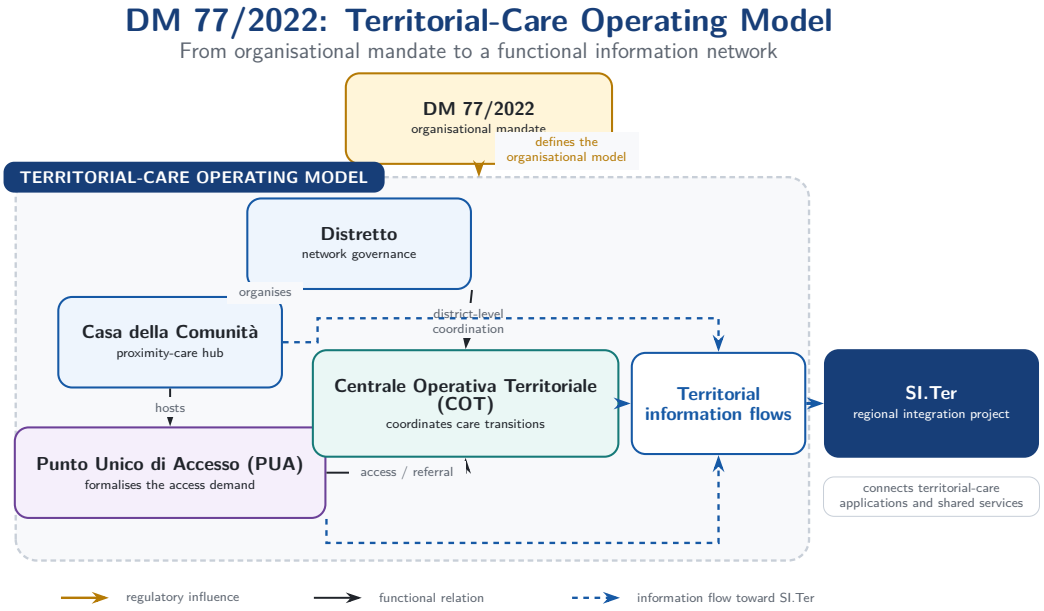

Figure 1.1: Concept of the territorial-care operating model defined by DM 77/2022.

This continuity requirement operates within a regulated data environment. Health data are a special category of personal data under Article 9 of European Union Regulation 2016/679, and their processing remains subject to data protection by design and by default and to measures appropriate to risk under Articles 25 and 32[3]. The FSE (Fascicolo Sanitario Elettronico), or Electronic Health Record, translates the same principle into role-based authorisation, periodic verification of access profiles, secure communication, operational traceability, and audit-log mechanisms[4]. DM 77/2022 thus creates a dual technical requirement: information must cross organisational boundaries to support the COT, yet each crossing must preserve identity, authorisation, confidentiality,

[3]European Parliament and Council of the European Union. *Regulation (EU) 2016/679 on the Protection of Natural Persons with Regard to the Processing of Personal Data and on the Free Movement of Such Data*. Regulation (EU) 2016/679. Official Journal of the European Union, L 119. European Union, Apr. 27, 2016, pp. 1–88. URL: https://eur-lex.europa.eu/eli/reg/2016/679/oj (visited on 07/29/2026), arts. 9, 25 and 32.

[4]Ministero della Salute. *Fascicolo sanitario elettronico 2.0*. Ministerial decree Decreto 7 settembre 2023. Italian Official Gazette, General Series no. 249, 24 October 2023, as subsequently amended. Ministero della Salute, Sept. 7, 2023. URL: https://www.gazzettaufficiale.it/eli/id/2023/10/24/23A05829/sg (visited on 07/29/2026), art. 25.

and traceability. Satisfying both conditions requires the territorial-care model to interoperate with the heterogeneous systems already present in the regional ecosystem.

1.1.2 Heterogeneous Systems in SI.Ter

The regional programme provides the Veneto project context in which the organisational mandate becomes an integration problem. Regional documentation defines SI.Ter as a programme that connects territorial-care application domains, shared services, and pre-existing local and regional systems; it does not assume their replacement by a single greenfield platform[5]. The procurement framework accordingly requires cooperation across distinct ownership boundaries while preserving the processes and information assets already managed by Healthcare Organisations[6]. The COT model therefore depends on an architecture whose components remain heterogeneous in responsibility, interface, and data representation. The integration perimeter includes APMS (Application Profile Management System) for authentication and user-profile management in cloud-based applications[7], Anagrafe Zero for patient identification and registration, local Admission, Discharge, and Transfer systems, and the FSEr (Fascicolo Sanitario Elettronico regionale) document infrastructure. Within the document path, XDS (Cross-Enterprise Document Sharing) repositories publish and retrieve clinical documents according to the regional Affinity Domain, while their indexing makes those documents available to the

[5]Regione del Veneto, Area Sanità e Sociale. *Modello di Governo per l'implementazione del Sistema Informativo Territoriale (SI.Ter) nelle Aziende ed Enti del Servizio Sanitario Regionale della Regione del Veneto.* Regional decree Decreto n. 11922. Regione del Veneto, Mar. 16, 2026, pp. 1–3.

[6]Azienda Zero. *Realizzazione di un Sistema Informativo Territoriale per gli Enti Sanitari della Regione Veneto.* Technical specifications. Specific procurement, Digital Healthcare – Clinical-Care Information Systems, ID 2365, Lot 3; date derived from file metadata. Azienda Zero, July 9, 2025; Raggruppamento Temporaneo di Imprese guidato da AlmavivA. *Sistema Informativo Territoriale per gli Enti Sanitari della Regione del Veneto – Lotto 3.* Technical report. Date derived from file metadata. Raggruppamento Temporaneo di Imprese guidato da AlmavivA, Sept. 19, 2025.

[7]DXC Technology. *Specifiche di integrazione autenticazione APMS.* Technical specification. Version 1.4.1. Azienda Zero, Nov. 22, 2024, Version 1.4.1, Section 1.

regional record. The documented SI.Ter architecture also connects context-call services, digital signature providers, prescription services, local clinical applications, and other shared components[8]. Within the integration stack represented by the thesis, Scryba Sign instantiates the signature-provider role and exposes its services through SOAP (Simple Object Access Protocol) interfaces described by WSDL (Web Services Description Language) contracts. The component handles FEA (Firma Elettronica Avanzata) and remote digital signatures; when the selected certificate and trust-service conditions satisfy the applicable eIDAS (electronic identification, authentication and trust services) requirements, the latter supports FEQ (Firma Elettronica Qualificata). It produces CAdES (CMS Advanced Electronic Signatures), PAdES (PDF Advanced Electronic Signatures), and XAdES (XML Advanced Electronic Signatures) envelopes and applies timestamping in the TimeStampedData (TSD) format. This classification identifies the supported integration surface and does not by itself certify the legal qualification of every configured signature[9]. The technological gap becomes visible in the interaction models exposed by the two principal data domains. Anagrafe Zero provides resource-oriented REST (Representational State Transfer) interactions over HTTP (Hypertext Transfer Protocol), based on HL7 (Health Level Seven) FHIR (Fast Healthcare Interoperability Resources) R4 (Release 4), for RVE (Regione Veneto) operations such as patient search and identifier assignment. FHIR admits both XML (Extensible Markup Language) and JSON (JavaScript Object Notation) representations, but the applicable regional interactions specify FHIR XML; the

[8]Raggruppamento Temporaneo di Imprese guidato da AlmavivA, *Sistema Informativo Territoriale per gli Enti Sanitari della Regione del Veneto – Lotto 3*, Sections 2.1.2 and 2.1.4.

[9]Medas S.r.l. *Scryba Sign 3.x Developer's Guide*. Developer's guide. Version 16. Restricted internal document. Medas S.r.l., Apr. 10, 2024, pp. 7, 10, 28, and 89; European Parliament and Council of the European Union. *Regulation (EU) No 910/2014 on Electronic Identification and Trust Services for Electronic Transactions in the Internal Market*. Consolidated text of 18 October 2024. July 23, 2014. URL: https://eur-lex.europa.eu/eli/reg/2014/910/2024-10-18/eng (visited on 08/05/2026), Articles 3 and 25; Adriano Santoni. *RFC 5544: Syntax for Binding Documents with Time-Stamps*. RFC Editor. Feb. 2010. DOI: 10.17487/RFC5544. URL: https://www.rfc-editor.org/info/rfc5544/ (visited on 08/05/2026), Section 2.

contrast is therefore not a simple XML-versus-JSON conversion[10]. The FSEr document infrastructure instead applies IHE (Integrating the Healthcare Enterprise) XDS.b transactions through SOAP (Simple Object Access Protocol) Web Services[11]. Both families can use HTTP over TCP (Transmission Control Protocol) and protect the channel through mTLS (mutual Transport Layer Security). The mediation challenge therefore lies above this common transport substrate: SI.Ter must reconcile resource-oriented REST operations with message-oriented SOAP transactions and map patient identifiers, document metadata, security contexts, error semantics, and workflow dependencies without changing their clinical meaning. Interface connectivity alone is insufficient because syntactically valid forwarding does not guarantee semantic equivalence between the two application contracts. An end-to-end care transition can consequently traverse several systems before the receiving actor obtains the information required to continue the process. The solution architecture mediates these interactions through application interfaces and an interoperability layer. The layer mediates RESTful FHIR and SOAP-based IHE XDS.b interactions with ebXML (Electronic Business using eXtensible Markup Language) metadata, while managing asynchronous orchestration, transformation, errors, and interface lifecycles[12]. Table 1.1 summarizes how the principal interoperability standards and regulatory constraints govern patient data, document exchange,

[10]Elena Vio et al. *Specifiche tecniche – Anagrafe Zero*. Technical specification. Version 2.6. Consorzio Arsenàl.IT, Jan. 9, 2024, Sections 1.5.2 and 2.5.2.

[11]IHE International. *IHE IT Infrastructure Technical Framework, Volume 1: Cross-Enterprise Document Sharing (XDS.b), Revision 20.2*. Nov. 11, 2025. URL: https://profiles.ihe.net/ITI/TF/Volume1/ch-10.html (visited on 07/22/2026); CNR (ICAR) e Dipartimento per la Trasformazione Digitale. *Specifiche tecniche per l'interoperabilità tra i sistemi regionali di FSE: Affinity Domain Italia*. Technical specification. Version 2.6.3. CNR (ICAR) e Dipartimento per la Trasformazione Digitale, Oct. 31, 2025.

[12]Raggruppamento Temporaneo di Imprese guidato da AlmavivA, *Sistema Informativo Territoriale per gli Enti Sanitari della Regione del Veneto – Lotto 3*, Sections 2.1.2 and 2.1.4.

access control, and auditability within the SI.Ter integration stack[13][14][15].

Table 1.1: Summary of key standards, including regulatory constraints, and their functional roles in the SI.Ter integration stack.

Standard	Protocol or format	Role in the project
HL7 FHIR R4 Health Level Seven Fast Healthcare Interoperability Resources, Release 4	HTTP/REST; `Patient` and `Bundle` resources represented in FHIR XML, with JSON where required by the applicable interaction	Supports patient-registry interactions with Anagrafe Zero, including patient search, identifier assignment, and demographic updates such as [RVE-54] and [RVE-55].
IHE XDS.b Integrating the Healthcare Enterprise Cross-Enterprise Document Sharing	SOAP/XML messages, ebXML metadata, and the ITI (IT Infrastructure) transactions ITI-41, ITI-18, and ITI-43	Supports publication, metadata indexing and query, and retrieval of clinical documents through the FSEr Document Repository and Document Registry.
SAML (Security Assertion Markup Language) 2.0 / JWT (JSON Web Token)	XML assertions and compact RFC (Request for Comments) 7519 tokens carried by the transaction-specific SOAP or HTTP/REST exchanges	Establishes and propagates the security context: [RVE-121] uses the SAML assertion to obtain a JWT, which is then used by [RVE-130] for the contextual service call.

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[13]European Parliament and Council of the European Union, *Regulation (EU) 2016/679 on the Protection of Natural Persons with Regard to the Processing of Personal Data and on the Free Movement of Such Data*, Article 25.

[14]Ministero della Salute, *Fascicolo sanitario elettronico 2.0*, Article 25.

[15]Azienda Zero, *Realizzazione di un Sistema Informativo Territoriale per gli Enti Sanitari della Regione Veneto*, Section 4.5.

Standard	Protocol or format	Role in the project
GDPR (General Data Protection Regulation) / FSE (Fascicolo Sanitario Elettronico) 2.0	Data Protection by Design and by Default constraints	Establishes design constraints for differentiated access profiles and operation traceability. In the regional stack, IHE ATNA (Audit Trail and Node Authentication) implements audit logging. Authorisation checks and audit-record generation constitute non-bypassable processing in a compliant path; their cost therefore remains part of the measured end-to-end performance.
eIDAS; CAdES, PAdES, and XAdES Electronic-signature legal framework and signature-envelope profiles	SOAP Web Services; CMS, PDF, and XML signature envelopes; TimeStampedData temporal evidence	eIDAS supplies the legal classification of electronic signatures, while the envelope and timestamp formats define the Scryba Sign integration artefacts processed by the calling application.

These capabilities solve the connectivity problem only at the interface level. They do not by themselves establish whether a multi-system care process remains reconstructable from its initial request to its final outcome. Each mediation function introduces a boundary at which identifiers, security contexts, payloads, or timing information may change. Authentication can be a prerequisite for a patient query; the patient identifier returned by that query can become an input to document search; a retrieved document can affect a later care decision. The integration gap is therefore not the absence of interfaces, but the absence of a single observation point that represents the meaning, ordering, and outcome of all interactions involved in a care workflow. This gap turns the heterogeneity of SI.Ter into a problem of information continuity and end-to-end traceability.

1.2 Continuity and Observability

1.2.1 Workflow Traceability

Territorial care crosses organisational, application, and information boundaries at the same time. Organisational fragmentation arises because a single care pathway assigns responsibilities to hospital departments, the COT, territorial services, general practitioners, and home-care providers. Application fragmentation arises because these actors use software governed by different organisations. Information fragmentation arises because identity, clinical, administrative, and workflow data reside in registries, local applications, repositories, and regional platforms. Interoperability connects these domains, but the care process remains traceable only when the connections preserve the relationships among actors, data, and ordered activities. The protected-discharge workflow illustrates the resulting dependency chain. A hospital operator identifies the patient, assembles the information required for the referral, selects a proposed territorial setting, and submits the request to the competent COT. The COT receives and evaluates the request, coordinates the actors involved, and advances the case through explicit workflow states. The protected-admission workflow applies an analogous coordination pattern in the opposite direction, from a territorial setting towards a hospital structure[16]. These processes are multi-actor workflows rather than sequences of unrelated service calls: their correctness depends on the availability and interpretation of outputs produced by earlier activities.

The upper branch in Figure 1.2 makes the identity dependency concrete. Within `FLOW_PATIENT_QUERY`, RDG executes [RVE-1.b] before the [RVE-54] Patient Query. Through the first transaction, the regional IAP (Identity and Assertion Provider) generates and returns a SAML 2.0 identity assertion; the execution engine extracts its required Base64 representation, stores it in the

[16]Raggruppamento Temporaneo di Imprese guidato da AlmavivA, *Sistema Informativo Territoriale per gli Enti Sanitari della Regione del Veneto – Lotto 3*, Sections 2.1.5 and 2.4.1.

context of the same flow occurrence, and resolves the corresponding placeholder in the `Authorization` header of [RVE-54] before sending the query. The resulting header carries the assertion with the `IHE-SAML` authentication scheme, as required by the regional security profile[17][18]. RDG therefore represents regional security-session state as explicit per-occurrence dependency data rather than reducing it to the state of the underlying HTTP connection. If [RVE-1.b] fails, the required assertion cannot be extracted, or its placeholder remains unresolved, the engine does not execute a semantically valid patient query and records the workflow as incomplete. This behaviour makes assertion acquisition and propagation part of the tested workflow. An isolated request can exercise [RVE-54] only after receiving a valid security context from outside the test and therefore cannot evaluate the identity hand-off on which the regional interaction depends.

A technical evaluation based on isolated endpoints cannot represent this dependency. A successful authentication request does not prove that the resulting security context is accepted by the next service; a successful patient query does not prove that the returned identifier reaches document search; and a set of individually successful calls does not prove that the workflow reaches its intended terminal state. The relevant unit of analysis is therefore one occurrence of an end-to-end workflow, considered both through its executed interactions and through its overall outcome and completion time. This unit preserves the functional meaning that disappears when measurements are aggregated only by endpoint. End-to-end reconstruction also requires stable correlation identifiers, ordered timestamps, transport and application outcomes, and explicit data dependencies between steps. Without this evidence, a failure can be located at the client boundary but cannot be associated reliably with the workflow state that precedes or follows it. The continuity problem consequently becomes an

[17] Mauro Zanardini et al. *Infrastruttura di sicurezza GDL-O Sicurezza.* Technical specification. Version 2.14. Consorzio Arsenàl.IT, June 28, 2024, Section 4.2.

[18] Vio et al., *Specifiche tecniche – Anagrafe Zero*, Section 1.7.

observability problem: the evaluation method must reconstruct the external behaviour of the distributed workflow, while the regional security perimeter limits which internal evidence the experiment can inspect.

Figure 1.2 maps this requirement to the Request Dataset Generator (RDG) flow model, in which each correlated occurrence retains both business outputs and the security context required by downstream transactions.

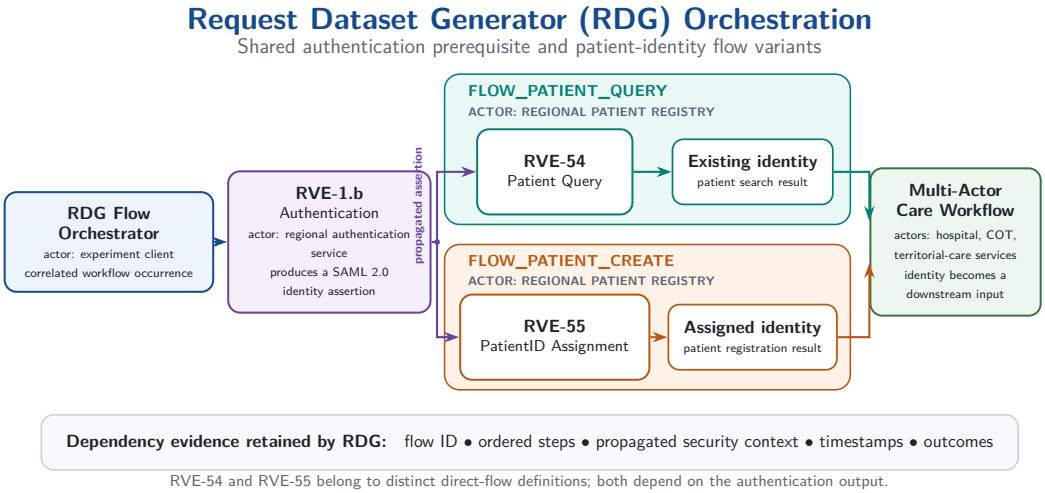

Figure 1.2: RDG orchestration of the [RVE-1.b] authentication output into regional patient-identity flows.

1.2.2 The Authorised Observation Boundary

Regional security specifications make authentication, authorisation, secure transport, and auditability integral to the integration architecture. Operators and applications act within an authenticated and authorised context, and assertions or tokens can become prerequisites for later services[19]. The architecture also provides audit mechanisms for production accountability; the regional security specification states that participating actors generate audit records and refers their detailed structure to the applicable ATNA (Audit

[19]DXC Technology, *Specifiche di integrazione autenticazione APMS*, Version 1.4.1; Zanardini et al., *Infrastruttura di sicurezza GDL-O Sicurezza*; Gregorio Canal and Gianluca Pavan. *Specifiche tecniche chiamata di contesto RVE*. Technical specification. Version 1.3. Consorzio Arsenàl.IT, July 18, 2025.

Trail and Node Authentication) profile[20]. The existence of this production telemetry does not make it part of the evidence available to every consumer or experiment, because access remains governed by operational roles and security policy. Within the authorised perimeter of this thesis, internal APMS telemetry and telemetry from the mediated gateway path are not made available to the experiment. This statement describes the research boundary; it does not claim that the platforms fail to produce logs or that regional policy universally prohibits their inspection. Real patient data, credentials, private endpoints, confidential deployment parameters, and production audit records likewise remain outside the disclosed dataset. Production services are neither instrumented by the thesis nor subjected to intrusive load.

These restrictions, summarised in Table 1.2, preserve the applicable security boundary but remove the internal timestamps required for a causal decomposition of gateway processing. The observable interval consequently extends from the client sending a request to the client receiving the corresponding response. It can include client-side processing, authentication, network transit, gateway mediation, downstream service execution, and response propagation. A latency measured across this interval characterises the path as a whole; it cannot attribute a delay to a specific gateway component without additional internal evidence. Treating the entire difference as gateway processing therefore exceeds the available data and violates the distinction between observation and causal explanation. This limitation determines the methodological requirement. The experiment needs a controlled external layer that represents complete workflows, fixes the workload and observation rules, records client-visible events, and compares direct with gateway-mediated execution at the same declared boundary. Such a layer does not replace production telemetry. It provides surrogate evidence for the effect visible outside the protected platform and makes every inference conditional on the comparability of the observed paths. The resulting research question is: *how can documented healthcare interop-*

[20]Zanardini et al., *Infrastruttura di sicurezza GDL-O Sicurezza*, Section 4.3.15.

erability workflows be translated into reproducible, observable, and measurable end-to-end workloads, and how can the apparent overhead associated with gateway mediation be estimated differentially when internal platform telemetry is outside the authorised observation perimeter?

Table 1.2: Comparison between internal production evidence and external experimental observation.

Dimension	**Authorised Observation Boundary (Internal)**	**Experimental Observation Perimeter (RDG)**
Data access	Real clinical data and ATNA audit records remain production-side and are accessible only to authorised operational roles.	RDG uses synthetic FHIR patients; real health data and production audit records remain outside the dataset.
Granularity	Component-level timestamps and traces remain protected; their availability depends on production instrumentation and authorised operator access.	RDG measures client-observed mediated-path latency $L_{\text{med}}(q)$ and the differential $\Delta L(q) = L_{\text{med}}(q) - L_{\text{dir}}(q)$ for the declared statistic q.
Outcome insight	Internal traces can associate a failure with a component only when the corresponding production evidence is available.	RDG distinguishes the HTTP transport outcome from application completion and semantic rejection visible in the response or output extraction.

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Dimension	**Authorised Observation Boundary (Internal)**	**Experimental Observation Perimeter (RDG)**
Telemetry	APMS, mediated-gateway, and production ATNA logs remain inaccessible to the experiment.	RDG produces correlated JSONL events for requests, responses, and completed workflow occurrences.
Overhead attribution	Direct gateway measurement requires aligned ingress, internal-component, and egress timestamps, which lie outside the experimental perimeter.	ΔL estimates an aggregate path-level effect that can include routing, security, transformation, downstream-service, and retry costs; it does not decompose APMS or gateway processing.

1.3 Methodological Response

1.3.1 Surrogate Observability through RDG

RDG answers this question as the technical instrument of a *Surrogate Observability Strategy*. The strategy translates documented healthcare interactions into synthetic and executable workloads, observes them at the client boundary, and compares controlled direct and gateway-mediated paths. Its central differential mechanism is the `direct_vs_middleware` experiment. The current implementation groups direct patient-query and patient-creation flows, contrasts their client-observed end-to-end completion-time distributions with the mediated patient-query flow, and reports differences for the mean and the 50th, 95th, and 99th percentiles. These differences estimate *apparent path-level overhead*; they do not decompose the internal execution time of the gateway.
Observation in this strategy is not synonymous with measuring network latency. The client-observed interval covers the complete request–response path, while RDG records the HTTP-level outcome and the application outcome as distinct evidence. An HTTP status code 200 confirms that the exchange returns a successful HTTP response, but it does not by itself prove that the payload contains the required output or that the workflow can advance. The execution engine therefore marks a step as failed when a mandatory output cannot be extracted, even if the response status is 200, and propagates that failure to the flow-level outcome. The same observation model retains service-reported semantic rejections, such as an [RVE-55] status code 422 associated with FHIR validation, without inferring the specific violated constraint when payload-level evidence is unavailable. The sanitised evidence set illustrates the first distinction through RVE-TOKEN responses that carry status code 200 but remain application failures because the required token is not extracted. [RVE-121] provides a complementary limit: its observed status code 200 demonstrates the current HTTP exchange, whereas evidence item `E-RVE-121-001` does not establish that a specific correction causes a transition from an earlier status code

400 to 200. Surrogate observability therefore separates HTTP-level evidence, application completion, and causal claims instead of treating every successful HTTP response as an equivalent logical outcome.

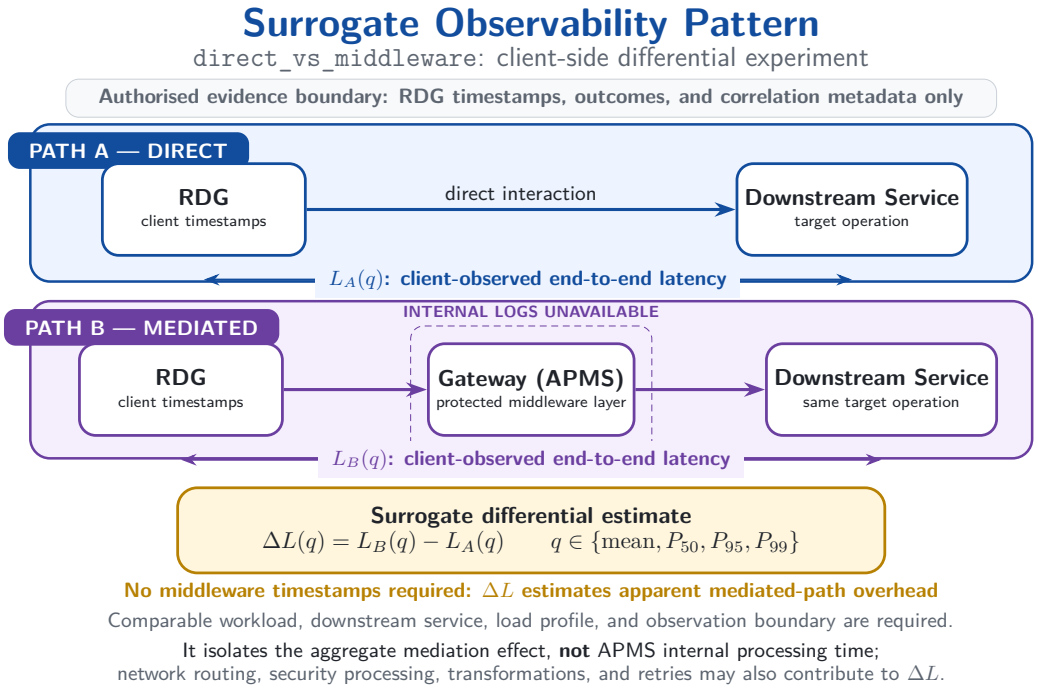


Figure 1.3: Surrogate observability pattern for `direct_vs_middleware` experiment.

The framework preserves the process semantics required by that comparison. It generates synthetic FHIR R4 patients, composes selected RVE (Regione Veneto) transactions into ordered multi-step flows, resolves outputs that become inputs to later steps, and assigns traffic occurrences to configurable timelines. Within each workflow occurrence, dependent steps execute sequentially; independent occurrences can execute concurrently through Python `asyncio`. Structured client-side events retain flow identity, step identity, timing, transport status, application outcome, and scheduling context. The resulting observation layer reconnects the functional workflow described by the SI.Ter documentation with the measurements available outside the protected gateway. The strategy covers selected workflow families related to regional authentication, patient identity, application context, document exchange, and digital

signature. Its standards perimeter includes HL7 (Health Level Seven) FHIR R4 resources, RVE specifications, IHE XDS.b transactions, SAML (Security Assertion Markup Language) assertions, and JWT (JSON Web Token) tokens. Synthetic patients replace real health information, while mock and authorised live modes separate controlled verification of the pipeline from observation of remote paths. This scope supports a reproducible performance method without claiming clinical effectiveness, legal certification, exhaustive coverage of SI.Ter, or successful validation of every implemented transaction.

1.3.2 Scope, Objectives, and Contributions

The method pursues four verifiable objectives:

1. interpret the standards, protocols, security mechanisms, and integration architectures that define the selected healthcare workflows;

2. formalise protected discharge, protected admission, and the associated interoperability interactions in terms of actors, preconditions, ordered steps, and information dependencies;

3. provide a configurable framework that translates those dependencies into synthetic, executable, and observable workloads;

4. define measurements that support differential comparison between direct and gateway-mediated paths.

The objectives establish an evidence-producing method and artefact; they neither promise a performance improvement nor predetermine the experimental outcome. RDG makes the objectives inspectable through a run-specific artefact chain. The chain comprises a resolved configuration, a synthetic dataset, a traffic timeline, a JSONL (JSON Lines) execution log, a normalised analytical dataset, metric reports, tabular summaries, and optional graphs. The analytical layer separates request latency, workflow completion time, throughput, error rate, concurrency, scheduling delay, and log completeness. Artefact presence

establishes software output coverage, whereas only an executed and quality-checked scenario provides experimental evidence. This distinction keeps the methodological contribution independent from any unsupported claim about observed performance. The contribution is consequently a reproducible procedure for profiling healthcare interoperability paths when the internal gateway remains outside the observation perimeter. The analytical component reconstructs the meaning and dependencies of the selected regional flows; the software component generates, executes, correlates, and measures their occurrences; the differential component quantifies the externally visible difference between declared direct and mediated paths. Together, these components close the causal chain that begins with the coordination mandate of DM 77/2022: the COT requires cross-setting information continuity, SI.Ter realises that requirement through heterogeneous integrations, distributed workflows create an end-to-end traceability problem, security constrains internal observation, and RDG supplies the surrogate evidence needed for controlled evaluation. The organisation of the thesis preserves the same progression. Chapter 2 examines the architectural, interoperability, workload, and performance foundations required to interpret the solution space. It also states the requirements, use cases, transactions, and ex ante evaluation criteria supported by the available sources. Chapter 3 presents RDG from the inside: framework design and implementation, data and workload generation, execution and observability, the experimental protocol, and the results produced by declared runs. Chapter 4 closes the traceability chain from objectives and artefacts to criteria, key performance indicators, and results; it assesses the contribution, states the limits of the evidence, and derives future directions from the actual findings.

Chapter 2

Solution Space

2.1 Scientific Basis of the Profiling Strategy

2.1.1 Experimental Alternatives and Validity

The methodological problem is not the generation of a large number of requests. It is the production of interpretable performance evidence for stateful healthcare workflows when the experimenter controls the initiating client but does not control the complete regional execution path. The relevant unit of analysis is therefore an information-dependent workflow rather than an isolated endpoint. A valid experiment preserves the order in which security material, patient identifiers, context information and application outcomes become available. It also distinguishes the load offered to the system from the work that the system completes.

Four experimental strategies define the solution space. Operational observation offers high ecological fidelity, but it does not isolate the offered load from concurrent changes in users, services, routing, and infrastructure. An isolated protocol benchmark offers strong repeatability, but it removes the state transitions that determine whether a regional workflow reaches its terminal outcome. Trace replay can preserve an observed temporal pattern, but it requires an authorised trace whose state, ordering, and dependencies support faithful reconstruction. Controlled synthetic workflows make composition, timing, concurrency, and randomisation explicit, but their resemblance to production traffic requires independent validation.

Zhu, Chen, and Chiueh provide the decisive methodological distinction. Their trace-replay study treats the input trace, the initial system state, and the dependencies among operations as necessary conditions for accurate replay. The authors also show that temporal or spatial scaling remains credible only when

it does not violate those dependencies[1]. Their result does not establish the universal superiority of replay over synthetic benchmarks. It establishes that replay validity depends on evidence that exists before the experiment begins. In the regional case examined by this thesis, no authorised production trace, complete state reconstruction, sanitisation procedure, or replay agreement enters the available evidence perimeter. Consequently, trace replay is not an admissible experimental option. Within this perimeter, controlled synthetic workflows constitute the only strategy that simultaneously preserves internal validity, data minimisation, and repeatability. The conclusion is conditional rather than absolute: synthetic execution is the only controllable route among the evidence sources that the experimenter is authorised to use. It does not reproduce an unknown operational distribution by assumption. Instead, it creates matched experimental populations whose differences follow from declared factors. This distinction prevents the absence of production evidence from becoming an implicit claim of production representativeness.

Table 2.1 separates the scientific role of each alternative from the claim that it can support.

[1]Ningning Zhu, Jiawu Chen, and Tzi-cker Chiueh. "TBBT: Scalable and Accurate Trace Replay for File Server Evaluation". In: *4th USENIX Conference on File and Storage Technologies (FAST 05)*. San Francisco, CA: USENIX Association, Dec. 2005. URL: https://www.usenix.org/conference/fast-05/tbbt-scalable-and-accurate-trace-replay-file-server-evaluation (visited on 07/29/2026).

Table 2.1: Experimental alternatives and the claims that each strategy can support within the available evidence perimeter. Source: Author's synthesis based on Zhu et al. (2005).

Strategy	**Primary strength**	**Admissible claim**
Operational observation	Natural service mix and environmental realism.	Descriptive evidence for the observed environment when access and provenance are complete.
Isolated protocol benchmark	Repeatable control of one exchange.	Baseline for one contract, not completion of a stateful healthcare flow.
Trace replay	Preservation of captured ordering and timing when state and dependencies remain reconstructable.	Reproduction of an authorised trace within its capture and scaling assumptions.
Synthetic workflow	Explicit flow mix, timing, seed, concurrency, and data policy.	Internally valid comparison of declared scenarios; external representativeness remains unproven.

The resulting validity model separates four questions. Construct validity concerns whether the executable workload preserves the documented meaning and prerequisites of each transaction. Internal validity concerns whether matched scenarios differ only in the intended factor. External validity concerns whether an independent regional source supports the configured temporal profile, payload classes, and flow mix. Conclusion validity concerns whether repeated observations, explicit populations, and visible missing data support the reported comparison. Reproducibility therefore does not imply representativeness, and a realistic observation does not imply causality when uncontrolled environmental changes coexist with the measured path.

2.1.2 Synthetic Workflows as Experimental Instruments

A synthetic workflow has scientific value only when it acts as an experimental instrument rather than as a collection of fabricated requests. Its structure derives from authoritative transaction and process specifications; its data remain synthetic; its schedule derives from declared stochastic or deterministic rules; and its outcomes remain classified at transport, application, and complete-flow levels. These properties make the independent variables inspectable and the dependent variables reproducible without requiring real patient records or security artefacts.

The strategy distinguishes *what* executes from *when* execution starts. The flow mix specifies semantically distinct workflows. The timeline assigns their planned start times. A homogeneous Poisson process supplies one reproducible baseline. For intensity $\lambda > 0$, the arrival count in an interval of duration t follows

$$\Pr[N(t) = k] = \frac{(\lambda t)^k e^{-\lambda t}}{k!}, \qquad k = 0, 1, 2, \ldots. \tag{2.1}$$

A non-stationary profile replaces the constant rate with $\lambda(t)$, for which

$$\mathbb{E}[N(a, b)] = \int_a^b \lambda(u)\, du. \tag{2.2}$$

Piecewise rates express controlled ramps and peaks, while parameterised bursts express short concentrations around a baseline. These models define experimental hypotheses; they do not estimate Veneto clinical demand in the absence of an authorised operational series.

The synthetic strategy preserves a second separation between flow arrivals and request arrivals. One flow can expand into different numbers of requests, carry different payload classes, and terminate at different points. Equal flow rates therefore do not imply equal request rates or equal remote work. A shift toward document processing changes transferred bytes and downstream computation; a shift toward authentication and context access changes the number of dependent steps and the probability of early termination. The experiment reports

both the configured flow composition and the realised transaction composition, because a load comparison without this information confounds intensity with workload content.

Synthetic data also perform a methodological role. Metadata-driven generation creates structurally and semantically valid patient and document contexts without using personal data. Stateful orchestration associates every derived identifier, assertion, token, or document reference with one flow occurrence, while a session-aware identity hand-off prevents security material from crossing concurrent occurrences. The asynchronous execution model permits concurrency among independent occurrences while preserving sequential dependencies inside each occurrence. These architectural properties support controlled concurrency without weakening the semantics of the workflow.

The design includes an explicit calibration condition. A controlled local path establishes the range in which scheduling, event production, and analysis sustain the intended offered load without material drift. This calibration does not predict the capacity of a regional service. It determines whether the workload generator remains a credible experimental instrument. When delivery drift already increases in the control condition, a corresponding live scenario cannot support a conclusion about external saturation.

The statistical unit follows the same hierarchy. Requests within one flow share state, flows within one run share workload and environmental conditions, and repetitions provide the highest-level comparable units. Request distributions remain useful for transaction analysis, but the method does not treat every request as an independent replicate. This rule prevents a large sample count from overstating the amount of independent experimental evidence.

The selected strategy consequently favours internal validity over unsupported realism. It makes every assumption visible: flow weights, arrival profile, payload class, connection policy, seed, concurrency bound, execution mode, and outcome rule. An independent authorised traffic source can later test external validity without changing the measurement semantics. Until such a source ex-

ists, the thesis presents synthetic workloads as controlled experimental inputs and never as reconstructed regional demand.

2.2 Security-Constrained Observation Strategy

2.2.1 Authorised Observation Boundary

The Authorised Observation Boundary originates from regional security, privacy, and organisational control, not from a limitation of the workload generator. The Request Dataset Generator (RDG) can initiate approved synthetic traffic and retain sanitised client-side events. The experimenter cannot inspect private execution phases inside the Application Profile Management System (APMS), regional middleware, or downstream services, and cannot retain assertions, access tokens, credentials, certificate secrets, private endpoints, or personal data. The boundary therefore determines which causal statements the evidence permits[2].

This constraint leads to an *Advanced Black-box with Surrogate Observability* methodology. The black-box component treats the protected regional path as an externally observable service. The advanced component preserves more than a request timestamp: it observes planned and actual dispatch, transport and application outcomes, dependency propagation, terminal flow state, flow completion time, scheduling drift, concurrency, and matched path differentials. The surrogate component uses these correlated external signals to characterise the path without pretending to expose its internal phases.

The boundary changes the meaning of overhead. A mediated request can include authentication, authorisation, envelope handling, semantic validation, transformation, routing, queueing, upstream waiting, and backend processing. A client-side duration contains their composite effect together with network and client contributions. It does not equal gateway processing time, and a slower mediated response does not identify one internal component as its cause.

[2]DXC Technology, *Specifiche di integrazione autenticazione APMS*; Zanardini et al., *Infrastruttura di sicurezza GDL-O Sicurezza*.

Optional authorised internal timestamps can validate or refine the external estimate, but their absence limits causal attribution rather than invalidating the matched external comparison. Figure 2.1 expresses this methodological boundary. The internal functions remain analytically plausible contributors, but the experiment measures only their aggregate path effect.

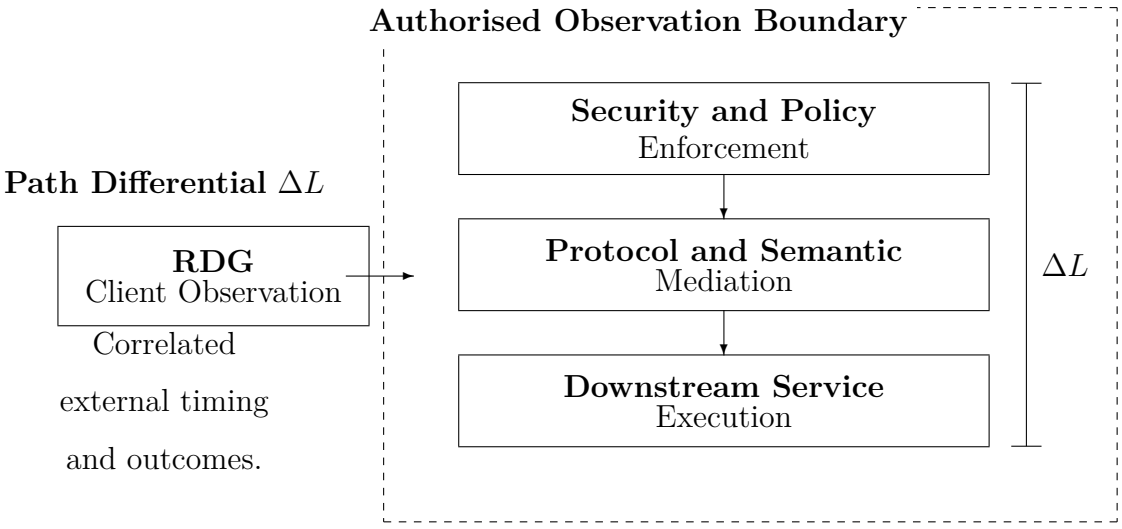


Figure 2.1: Security-constrained surrogate observability. The visual defines the limit of causal attribution adopted by the thesis. Source: Author's elaboration based on DXC Technology (2024) and Zanardini et al. (2024).

Security state remains part of the workload even when its value remains absent from the evidence. Reusing one assertion or token across unrelated occurrences can increase apparent throughput while violating the transaction contract and removing a mandatory acquisition cost. The method therefore scopes security outputs to one occurrence and records only whether their production and propagation succeed. Data minimisation removes protected values, not the causal position of the security transition.

Mock middleware events serve a separate purpose. They validate parsing, correlation, missing-field handling, and metric computation. They do not support claims about production routing, policy enforcement, queueing, or backend processing. This evidence separation prevents a test of the analysis chain from

becoming surrogate evidence about the regional infrastructure.

2.2.2 Client-Boundary Latency and Saturation Evidence

Iorio, Risso, and Casetti justify the choice of the client boundary by distinguishing network round-trip time from application-level flow completion time. Their end-to-end interval begins with request transmission and ends with complete response reception; it therefore captures transmission, transport, middlebox, application-protocol, and service contributions that a network probe does not expose[3]. This boundary also avoids the clock-synchronisation requirement of one-way measurements. For a protected regional path, the same principle makes the client the only common observation point across direct and mediated scenarios.

The full-stack interpretation makes Flow Completion Time (FCT) the principal end-to-end timing metric. Iorio et al. show that network round-trip time (RTT) captures only one layer of a latency budget that also contains transmission, forwarding, queueing, transport establishment, application protocol, security, service exposure, orchestration, and computation. RDG extends this observation boundary from one request-response exchange to the terminal outcome of an information-dependent workflow. Request latency remains necessary for locating which transaction population changes, but workflow FCT determines when the domain operation actually completes. This extension transfers the measurement principle, not the numerical results of the edge-computing campaign[4].

Figure 2.2 makes the measurement hierarchy explicit. The horizontal displacement between the network, transport, and application distributions represents progressively accumulated work rather than three interchangeable latency measures. The curves provide a conceptual reading of the hierarchy analysed by

[3]Marco Iorio, Fulvio Risso, and Claudio Casetti. "When Latency Matters: Measurements and Lessons Learned". In: *ACM SIGCOMM Computer Communication Review* 51.4 (Oct. 2021), pp. 2–13. DOI: 10.1145/3503954.3503956.

[4]Ibid.

Iorio et al.

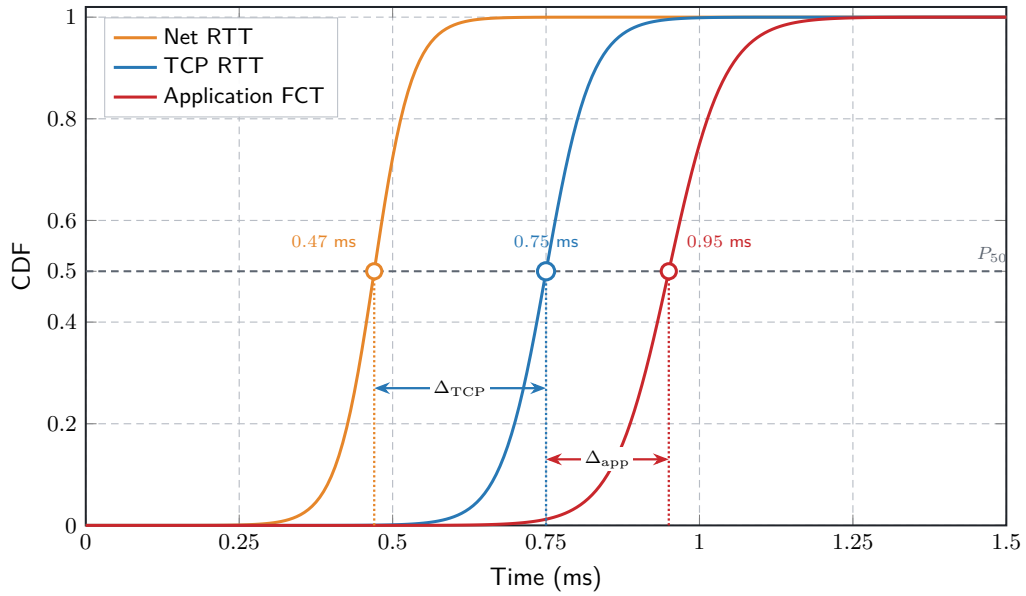


Figure 2.2: Conceptual separation of Network Round-Trip Time (RTT), Transmission Control Protocol (TCP) RTT, and application-level Flow Completion Time (FCT). Source: Adapted from Iorio et al. (2021).

For completed request i, the external latency is

$$L_i = t_i^{\text{response}} - t_i^{\text{dispatch}}. \tag{2.3}$$

For matched direct and mediated populations, the primary path estimand is

$$\Delta L = \widetilde{L}_{\text{mediated}} - \widetilde{L}_{\text{direct}}, \tag{2.4}$$

where $\widetilde{L}$ denotes a declared statistic over compatible transaction, payload, outcome, connection-policy, load, and time-window classes. The differential estimates the external effect associated with the path change. It does not decompose the contribution of APMS, transport, protocol transformation, or a downstream service.

Consequently, ΔL represents a *Path-level Overhead*: an aggregated external differential that can contain propagation, transport and security establishment,

gateway policy enforcement, SOAP/XML or REST/JSON handling, semantic validation, transformation, routing, queueing, and downstream service time. The term *aggregated* is essential. A positive differential supports the claim that the mediated path adds externally observable time under matched conditions; it does not identify which protected internal phase causes that difference. This interpretation follows the multi-layer latency budget in the edge-computing literature while respecting the narrower authorised observation boundary of the regional case.

The client boundary becomes informative about saturation only when latency joins other signals. Akram, and Rehman model distributed load sharing as a race between communication delay and a stochastic availability window. Their analysis links increasing traffic intensity and communication latency to stale state, failed transfers, wasted work, and reduced effective throughput[5]. The load-sharing model does not become a model of the Veneto architecture, and its numerical thresholds do not transfer to RVE transactions. Its methodological contribution is the joint interpretation of latency, state validity, and completed useful work: delay becomes a saturation signal when it coexists with shrinking completion capacity and increasing ineffective activity.

Figures 2.3 and 2.4 translate this contribution into two complementary views. The state model distinguishes useful transitions from a transfer that arrives after the target state loses validity. The distribution view shows the same risk as a narrowing Load-Sharing Window (LSW) when traffic intensity increases. Both views remain methodological abstractions and import neither the paper's numerical thresholds nor a topology for the regional system.

[5]Saira Akram, Muhammad Asim Akram, and Fattah UR Rehman. "Numerical Evaluation of Network Latency and Throughput in Distributed Systems". In: *The Science Post* 1.4 (Dec. 2025). DOI: 10.5281/zenodo.18012513.

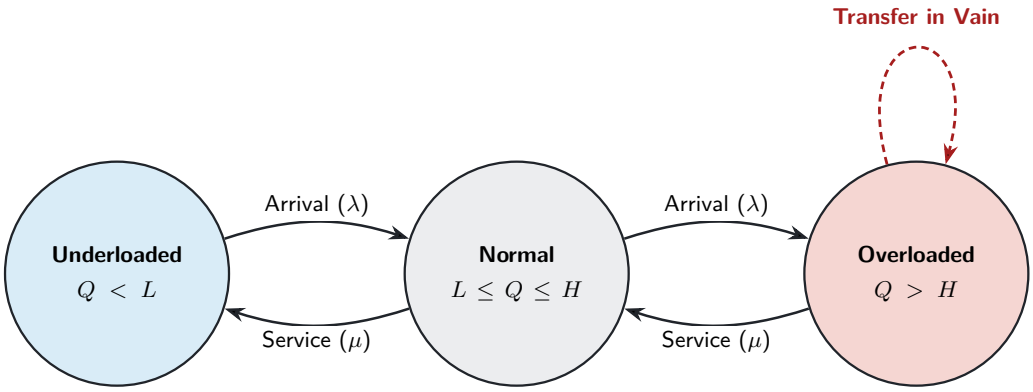


Figure 2.3: Conceptual transitions between underloaded, normal, and overloaded node states. Source: Adapted from Akram et al. (2025).

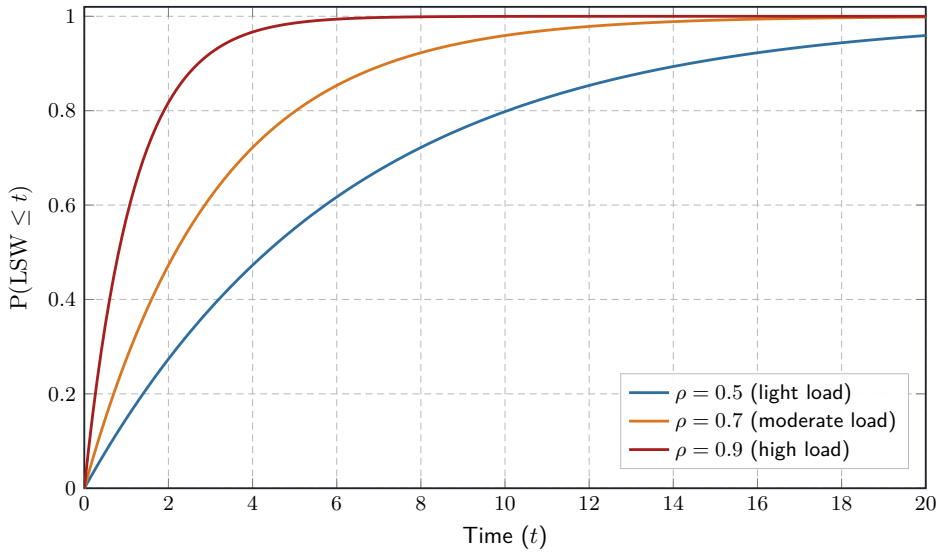


Figure 2.4: Conceptual cumulative distribution of the Load-Sharing Window under increasing traffic intensity. Source: Adapted from Akram et al. (2025).

This stochastic interpretation determines the summary statistics. The median (P_{50}) supplies a robust centre for a skewed latency population, while P_{95} and P_{99} expose the tail in which a delayed prerequisite is most likely to outlive the state that makes it useful. The arithmetic mean remains a supplementary descriptor, but it cannot reveal the probability of crossing a high-latency boundary and it changes markedly under rare delays. Akram et al. justify this distribution-aware choice through a quantile rule rather than through average

latency alone; Iorio et al. likewise evaluate application FCT at high percentiles. RDG therefore reports P_{50}, P_{95}, and P_{99} for a declared homogeneous population and observation window, together with sample size and outcome coverage. The Load-Sharing Window does not represent an actual SI.Ter component: it supplies a conceptual model for the stochastic interval in which information about distributed state remains actionable[6].

Dimitrov and Skjellum reinforce the need for application-level interpretation. Their experiments show that lower point-to-point latency does not necessarily produce lower application execution time, because communication pattern, message size, completion mode, CPU overhead, and the possibility of overlapping work determine the complete effect[7]. The studied Message Passing Interface (MPI) environment differs from healthcare interoperability, so the thesis transfers only the evaluation principle: a local latency number cannot stand for the performance of a heterogeneous workflow. Healthcare applications make this principle especially relevant because patient queries, context transitions, document retrieval, and cryptographic signing expose different message sizes, synchronisation points, and blocking dependencies.

An isolated ping-pong benchmark therefore answers a narrower question than the one posed by regional mediation. Dimitrov and Skjellum show that a lower point-to-point latency can coexist with higher processor overhead or reduced opportunity to overlap communication and computation; the regular request-reply pattern also omits the collective and asynchronous structures of real applications. In the healthcare case, an isolated endpoint call omits semantic validation, identity acquisition, branch selection, and failure propagation. RDG is consequently necessary as an orchestrator of dependent workflows, be-

[6] Akram, Akram, and Rehman, "Numerical Evaluation of Network Latency and Throughput in Distributed Systems"; Iorio, Risso, and Casetti, "When Latency Matters: Measurements and Lessons Learned".

[7] Rossen Dimitrov and Anthony Skjellum. "Impact of Latency on Applications' Performance". Manuscript consulted in the documentary corpus. Apr. 2001. URL: https://users.cs.northwestern.edu/~srg/Papers/01-10-02/srg_2.pdf (visited on 07/31/2026).

cause the critical path exists only when each valid output enables the next step.[8].

Figure 2.5 provides an explanatory adaptation of the point-to-point and collective categories in their communication-pattern analysis. The original figures report message counts and sizes rather than sequence traces; the visualisation therefore illustrates the structural distinction without claiming to reconstruct an observed execution.

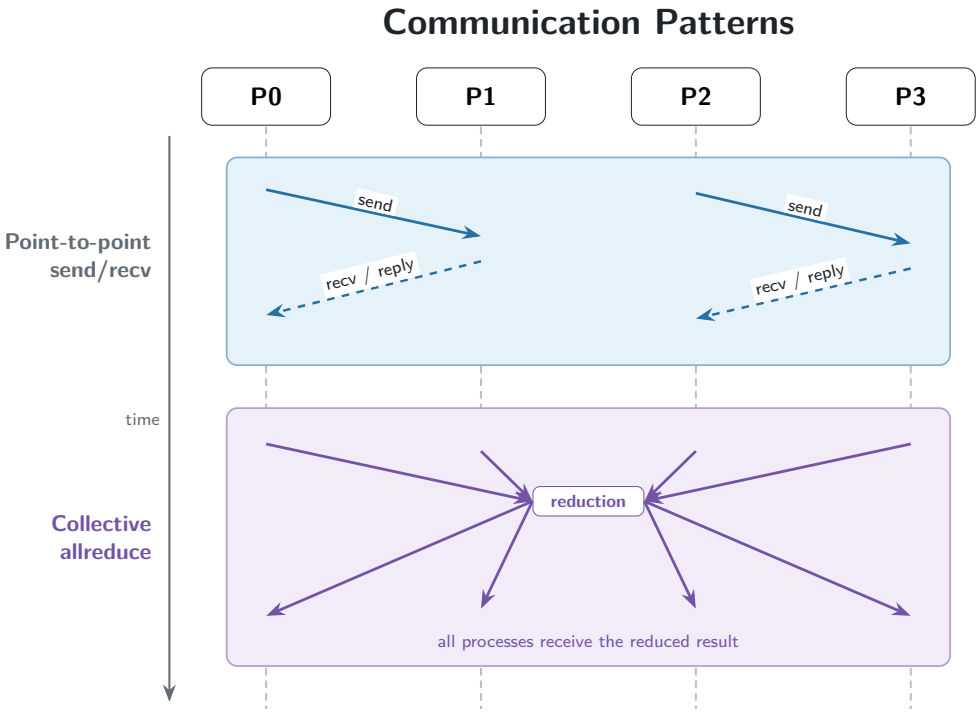


Figure 2.5: Conceptual point-to-point send/receive and collective all-reduce communication patterns. Source: Adapted from Dimitrov and Skjellum (2001).

Latency variability matters for the same reason. A median describes the centre of one population, but a dependent flow reacts to the slowest prerequisite on its critical path. Iorio et al. document dispersion, high-percentile instability, and latency spikes across access, transport, and application conditions; Dimitrov and Skjellum show that the application effect depends on the sur-

[8]Dimitrov and Skjellum, "Impact of Latency on Applications' Performance".

rounding communication pattern. In a healthcare workflow, variability can delay the availability of a patient identity, context token, document reference, or signed artefact even when the median request remains stable. The method therefore reports distributions and tails for homogeneous populations and never substitutes a point-to-point mean for complete-flow behaviour.

Saturation is consequently an inference from converging evidence rather than a synonym for high latency. The method requires persistent scheduling drift, occupancy near the configured concurrency bound, a plateau or decline in useful throughput, and a corresponding change in latency, timeouts, or semantic failures. Stable response latency can coexist with increasing delivery drift when the generator no longer dispatches work on time. Conversely, higher latency without a throughput plateau can reflect payload composition, connection establishment, or remote environmental variation rather than saturation.

Table 2.2 fixes the operational meaning of the principal signals.

Table 2.2: Operational measures and their role in saturation analysis. Source: Author's synthesis based on Iorio et al. (2021) and Akram et al. (2025).

Measure	**Population and unit**	**Interpretive role**
Request latency	Homogeneous completed attempts; milliseconds.	Client-observed dispatch-to-response distribution for one transaction class, reported through P_{50}, P_{95}, and P_{99} with the sample size.
Flow completion	Complete and interrupted occurrences remain separate; milliseconds.	Critical-path effect of orchestration, waiting, and blocking failures.
Useful throughput	Successful terminal flows per declared window.	Completed domain work; it remains distinct from raw request activity.
Request throughput	Completed attempts per declared window.	Technical activity, including calls that belong to unsuccessful flows.
Scheduling drift	Actual minus planned start; milliseconds.	Delivery quality of the generator and early evidence of backpressure.
Outcome distribution	Transport, application, timeout, and terminal-flow classes.	Reliability and preservation of unsuccessful work in every denominator.

The literature therefore supports the measurement boundary and the multi-indicator reasoning, not an imported acceptance threshold. Client-boundary latency supplies a reproducible external signal; ΔL controls the matched path comparison; throughput, drift, concurrency, and outcomes determine whether the signal is consistent with saturation. Internal attribution remains outside the authorised boundary unless a new evidence source enters the protocol.

2.3 Healthcare Workflows as Information Dependencies

2.3.1 Protected Discharge and Anagrafe Zero

Protected discharge gives clinical-process meaning to the profiling strategy. The inpatient unit identifies the patient, prepares assessment information, proposes a destination setting, and forwards the request to the Centrale Operativa Territoriale (COT). The COT evaluates the request and assigns the definitive setting[9][10]. Each stage consumes information that a preceding stage makes available. The workflow is therefore a chain of semantic prerequisites, not a bag of independent service calls.

Patient identity illustrates the first dependency. The project documentation associates protected discharge with patient identification through Anagrafe Zero, while the Anagrafe Zero specification requires [RVE-54] Patient Query before other registry transactions[11]. A successful network response alone does not satisfy this prerequisite. The query has to produce an application outcome and a patient context that the next branch can use. Depending on the profiled outcome, a later operation can assign a new regional identity through [RVE-55] or update an existing identity through [RVE-57]. The branch follows the semantic result of the query; it does not follow only the HTTP status.

This dependency explains why a session-aware orchestrator is necessary. The patient context belongs to one synthetic occurrence and cannot leak into another concurrent occurrence. A failed query blocks the dependent branch, and a warning or application error remains distinguishable from transport success. A generic request generator can reproduce this behaviour only after the test author adds domain-specific state, extraction, validation, and branch rules. The profiling problem therefore resides in semantic continuity, not in the abil-

[9] Raggruppamento Temporaneo di Imprese guidato da AlmavivA, *Sistema Informativo Territoriale per gli Enti Sanitari della Regione del Veneto – Lotto 3*, pp. 18–21.

[10] Azienda Zero, *Realizzazione di un Sistema Informativo Territoriale per gli Enti Sanitari della Regione Veneto*, pp. 35–36.

[11] Vio et al., *Specifiche tecniche – Anagrafe Zero*.

ity to send a HL7-FHIR R4 request. Semantic FHIR R4 generation is therefore a condition of construct validity, not a convenience for producing payload volume. Anagrafe Zero interprets identifiers, demographic search parameters, resource structure, and transaction outcomes according to its declared FHIR contract. A syntactically valid but semantically arbitrary JSON or XML document can exercise transport while triggering a different validation branch from the represented patient operation. Metadata-driven generation preserves the profile-relevant structure and makes the resulting query, assignment, or update comparable across repetitions. It also keeps the data synthetic, so semantic fidelity does not require personal records[12].

The same principle applies to regional security and context access. An [RVE-1.b] interaction produces a SAML assertion; [RVE-121] uses the assertion and contextual information to produce the JWT required by [RVE-130][13]. The later call remains invalid when the required transition is absent, even if an earlier endpoint returns a nominally successful transport status. Occurrence-local state preserves both the security contract and the cost of acquiring that state.

Figure 2.6 makes the occurrence-local dependencies visible. It keeps the context path and the patient-query path separate: [RVE-130] consumes the material produced through [RVE-121], whereas [RVE-54] consumes the assertion in a distinct FHIR query. This separation prevents the visual model from suggesting one unsupported SAML–JWT–FHIR transaction chain.

[12] Vio et al., *Specifiche tecniche – Anagrafe Zero*.

[13] Zanardini et al., *Infrastruttura di sicurezza GDL-O Sicurezza*; Canal and Pavan, *Specifiche tecniche chiamata di contesto RVE*.

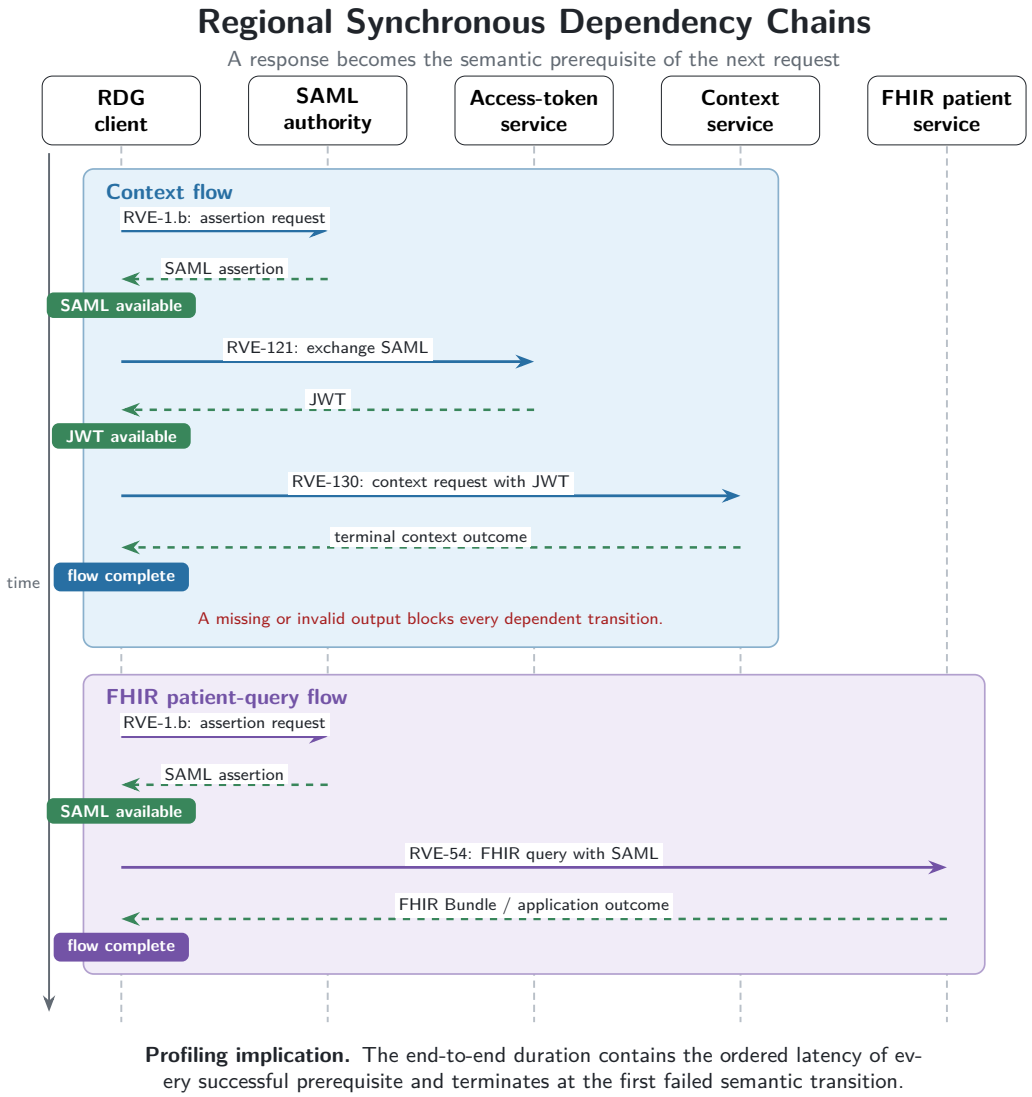


Figure 2.6: Conceptual regional dependency chains retained by the workload model. Source: Author's elaboration based on the regional security specification v2.14 (2024), context-call specification v1.3 (2025), and Anagrafe Zero specification v2.6 (2024).

Document access adds a further dependency family. Integrating the Healthcare Enterprise (IHE) Cross-Enterprise Document Sharing (XDS.b) separates metadata query, document selection, retrieval, and publication[14]. An Information

[14]IHE International, *IHE IT Infrastructure Technical Framework, Volume 1: Cross-Enterprise Document Sharing (XDS.b), Revision 20.2*; CNR (ICAR) e Dipartimento per la Trasformazione Digitale, *Specifiche tecniche per l'interoperabilità tra i sistemi regionali di FSE: Affinity Domain Italia*.

Technology Infrastructure ITI-18 query returns metadata; an ITI-43 retrieval consumes a repository and document reference; an ITI-41 publication carries document content and metadata. These transactions remain separate performance populations because they differ in payload, outcome, and position in the information chain.

Table 2.3 summarises the dependencies that shape the workload model without claiming that every listed transaction forms one approved protected-discharge implementation.

Table 2.3: Information dependencies retained by the profiling model. Source: Author's synthesis of the Anagrafe Zero v2.6 (2024), regional security v2.14 (2024), context-call v1.3 (2025), Affinity Domain Italia v2.6.3 (2025), Scryba Sign v16 (2024), and SI.Ter (2025) technical specifications.

Dependency	**Semantic prerequisite**	**Profiling consequence**
Patient identity	[RVE-54] produces a usable registry outcome before assignment or update.	The flow branches on application semantics and keeps later steps blocked after failure.
Security identity	[RVE-1.b] produces occurrence-local assertion evidence.	Protected calls include acquisition cost without retaining the assertion value.
Context session	[RVE-121] produces the material required by [RVE-130].	Transport success without the required output remains a failed state transition.
Document access	ITI-18 produces the reference consumed by ITI-43.	Query and retrieval retain separate latency, payload, and outcome populations.
Care transition	Patient identity and assessment precede COT evaluation and destination assignment.	Technical completion does not imply clinical approval or completion of the care pathway.
Document Signing	SignOneDoc transaction need a SAML Assetion Token from a [RVE-1.b].	Transport success without the required output remains a failed state transition.

Both care transitions act as process frames. They identify the initiating actor, decision point, information prerequisites, and terminal organisational outcome. They do not justify an invented chain among every available RVE and ITI transaction. A transaction enters an executable case only when both a technical

specification and a process source support its role. This source gate protects the profiling model from turning architectural proximity into an unsupported workflow dependency.

Anagrafe Zero also demonstrates why throughput requires a domain denominator. Three successful endpoint responses do not necessarily represent three completed patient-identity processes. A query can end the flow, enable an assignment, or enable an update; warnings and duplicate-management rules can change the terminal meaning. Request throughput therefore measures technical activity, while flow throughput measures useful completion under an explicit semantic outcome rule.

2.3.2 Cryptographic and Protocol Heterogeneity

Scryba Sign supplies a controlled example of heterogeneous interoperability. The service surface uses SOAP, XML, and WSDL. contracts over protected transport. The synchronous path combines identity and signer discovery through *GetUserInfo4* with the *SignOneDoc* operation. The document can travel as a Base64-encoded payload, and the service performs cryptographic work and constructs a signed artefact[15]. Scryba Sign therefore represents a heterogeneous interoperability constraint, not an optional software module. Treating it as an ordinary document upload removes the protocol, identity, and cryptographic work that the profile has to observe.

The methodological question is why the signing path enters the workload, not how the client constructs a particular envelope. A realistic profile has to retain the payload class, signing mode, session semantics, and terminal outcome because each factor changes the critical path. SOAP/XML handling adds representation and envelope costs; Base64 increases the transferred representation relative to the binary source; protected transport adds connection and security work; cryptographic signing adds service-side computation. The Authorised Observation Boundary makes only their aggregate external effect observable as

[15]Medas S.r.l., *Scryba Sign 3.x Developer's Guide*, pp. 17–18, 91–96, 107–111.

part of the Path-level Overhead. Signer identity creates a further dependency. The information-discovery step identifies the available signing powers, certificates, and One-Time Password (OTP) devices; the signing request depends on an admissible certificate and on the applicable OTP authorisation path[16]. This relationship requires session-aware identity hand-off: certificate selection and One-Time Password state belong to the same signing occurrence and a failed identity prerequisite blocks the cryptographic operation. The complete FCT consequently includes the ordered identity and signing path rather than only the final SOAP response. The current executable perimeter preserves the order *GetUserInfo4–SignOneDoc* for the synchronous population; automated certificate selection and the complete OTP lifecycle remain contract-level extensions, so the thesis does not present them as executed measurements.

The synchronous and asynchronous scenarios define distinct experimental populations. In the synchronous scenario, the request-response chain includes the return of the signed document. In the asynchronous scenario, submission produces session and document identifiers, while later status checks and retrieval define completion. The asynchronous path therefore introduces queue residence, polling or notification policy, and a longer state lifecycle[17]. Pooling the two modes in one latency distribution destroys construct validity because their terminal events and critical paths differ.

The profiling model consequently stratifies signing evidence by execution mode, payload class, connection policy, and application outcome. The current executable evidence perimeter includes the synchronous signing population. The asynchronous contract remains a separately defined profile and does not acquire results by analogy. This boundary preserves architectural completeness without presenting an unexecuted scenario as observed evidence.

[16] Medas S.r.l., *Scryba Sign 3.x Developer's Guide*, pp. 60–62, 91–96.

[17] Ibid., pp. 17–18.

Table 2.4: Methodological separation of Scryba Sign profiling populations; the distinct terminal conditions prevent synchronous, asynchronous, and large-document paths from being pooled into one latency distribution. Source: Author's synthesis based on the Scryba Sign 3.x Developer's Guide v16 (Medas S.r.l., 2024).

Profile	**Terminal condition**	**Relevant overhead**
Synchronous signing	The signed artefact returns in the synchronous interaction.	SOAP/XML and Base64 transfer, protected transport, cryptographic processing, and response transfer.
Asynchronous signing	Submission state leads to later availability and retrieval of the signed artefact.	Submission, queue residence, status acquisition, retrieval policy, and state lifecycle.
Large-document path	The signed artefact completes through the applicable document-transfer procedure.	Document size, hashing, external transfer, signing, and retrieval remain explicit strata.

A direct-versus-mediated comparison uses the same signing mode, payload class, application outcome, and connection policy on both paths. Under these controls, ΔL estimates the composite external effect associated with mediation. It does not isolate cryptographic time, SOAP serialisation, policy enforcement, or network transfer. The signing service therefore strengthens the need for surrogate observability: it creates a relevant, heterogeneous critical path while leaving its protected internal decomposition unavailable.

The scenario also confirms the importance of complete-flow metrics. A fast submission does not imply a fast signed-document outcome, just as a successful patient query does not imply completion of a protected-discharge process. The method keeps request latency, state-transition success, and terminal flow

completion distinct so that local progress cannot masquerade as domain-level completion.

2.4 Experimental Design and Evidence Semantics

2.4.1 Controls, Variables, and Reproducibility

The primary estimand is the change that RDG observes when a documented operation follows a mediated path instead of a direct path under matched conditions. The independent variable is the selected path. Transaction class, payload class, semantic outcome rule, flow position, offered-load profile, connection policy, concurrency bound, seed policy, duration, and observation window act as controlled variables. Request latency, flow completion, useful throughput, scheduling drift, and outcome distribution act as dependent variables.

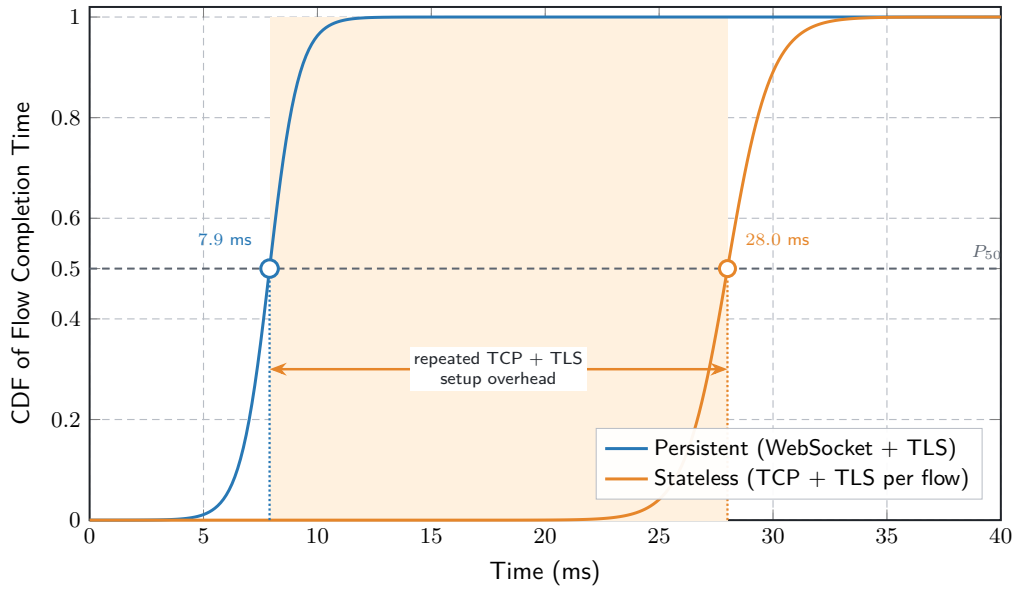
Matching requires more than a shared transaction identifier. Direct and mediated populations use compatible payloads, success definitions, flow positions, and workload conditions. Scenario order also remains controlled because connection reuse, cache state, or remote environmental drift can create an apparent path difference. Paired seeds align generated choices, while deterministic shuffling or interleaving reduces a fixed chronology effect. The method retains run timestamps and dispersion because a live remote environment remains variable even when the synthetic workload is reproducible.

Table 2.5 identifies the main confounding sources and the residual limitations that remain after control.

Table 2.5: Controls for an interpretable direct-versus-mediated differential. Source: Author's synthesis based on Zhu et al. (2005) and Iorio et al. (2021).

Confounding source	**Experimental control**	**Residual limitation**
Transaction semantics	Match operation, flow position, and application outcome.	Contract evolution can change equivalence across versions.
Payload and connection state	Stratify payload class and declare connection policy.	Private caches and remote connection management remain opaque.
Offered load	Reuse temporal profile, mix, seed policy, duration, and concurrency.	Unobserved external demand remains uncontrolled.
Execution chronology	Pair repetitions and control scenario order.	Environmental drift can persist within and across pairs.
Generator capacity	Calibrate delivery and retain drift and active-concurrency signals.	Host scheduling remains part of the external experiment.

Every run identifies the workload profile, offered rate, duration, seed, burst parameters, flow mix, concurrency bound, execution mode, configuration state, and artefact version. Reusing a seed aligns paired choices; changing seeds across repetitions samples alternative schedules. Neither action makes the remote environment deterministic. Reproducibility refers to the experimental input and analysis procedure, while environmental repeatability remains an empirical question. Connection policy deserves explicit control because security setup can dominate short flows. Figure 2.7 gives a conceptual reconstruction of the persistent and stateless populations analysed.



RDG design implication. Preserve occurrence-local security state and reuse transport sessions whenever the transaction contract and the authorised security policy permit it.

Figure 2.7: Conceptual comparison of Flow Completion Time (FCT) under persistent and stateless secure-connection policies. Source: Adapted from Iorio et al. (2021).

Durations use a monotonic clock. Wall-clock timestamps establish chronology but do not support subtraction across independent hosts without a documented synchronisation bound. Correlation requires compatible identifiers, event semantics, and populations. Missing or unmatched fields remain absent instead of being inferred. These rules prevent timestamp precision from being confused with visibility beyond the authorised boundary.

Structured events retain run, scenario, occurrence, flow, step, and request identity together with planned time, measured duration, and outcome class. Normalisation preserves malformed records, missing terminal events, interrupted flows, and unmatched correlations as data-quality outcomes. Silent removal of these records biases both latency and reliability toward successful observations. Provenance therefore precedes performance interpretation.

2.4.2 Evidence Gates and Ex Ante Judgement

Every performance statement requires an operational definition, a population, a unit, an aggregation level, and an event source. If $C(w)$ denotes the number of completed units in a window w, then

$$X(w) = \frac{C(w)}{|w|} \tag{2.5}$$

defines throughput only after the method declares whether $C(w)$ counts requests, terminal flows, or successful terminal flows. Retries can increase request throughput without increasing useful flow throughput, and early failures can reduce request activity while increasing the number of incomplete flows.
Criteria remain fixed before the corresponding results enter the judgement. For criterion j, the thesis uses

$$C_j = (S_j, P_j, m_j, s_j, w_j, d_j, \tau_j, E_j), \tag{2.6}$$

where S_j is the scenario, P_j the population, m_j the metric, s_j the statistic, w_j the window, d_j the preferred direction, τ_j an externally justified threshold, and E_j the evidence gate. A numerical value is not assessable when its population, boundary, or threshold provenance is missing.

Table 2.6: Ex ante criteria and required evidence. Source: Author's methodological synthesis based on the SI.Ter procurement specification (Azienda Zero, 2025).

Criterion	**Population**	**Required evidence**
Latency	Homogeneous requests and complete or interrupted flows remain separate.	Timer boundary, statistic, sample size, window, repetitions, path, and outcome rule.
Throughput	Request and useful-flow completions under declared offered load.	Profile, mix, concurrency, calibration, duration, and plateau rule.
Reliability	Transport, application, timeout, and terminal-flow outcomes.	Error taxonomy, denominator, retry rule, and semantic outcome rule.
Saturation	Drift, active concurrency, throughput, latency, and failures.	Repeated convergence of indicators rather than offered rate alone.
Data quality	Expected, parsed, malformed, incomplete, and unmatched events.	Event semantics, provenance, coverage rule, and completeness threshold.

Observation completeness conditions every statistic. The configured timeline and flow definition determine expected requests and terminal events; the retained event stream determines parsed observations. A missing end event, duplicate identifier, malformed record, or unmatched occurrence changes the admissible population. Successful-request latency therefore appears beside timeouts and application failures rather than replacing them.

The judgement vocabulary follows the evidence hierarchy. A criterion is *not assessable* when its threshold or evidence gate is absent; *conditionally supported* when structural or controlled evidence exists without the required live coverage; *accepted* when valid evidence meets an externally declared rule; and *rejected*

when valid evidence violates that rule. Mock evidence supports instrument validation. Live external evidence supports observations for the declared environment. Internal causal evidence requires a newly authorised source inside the protected boundary.

Threshold provenance remains external to the campaign under judgement. An approved service-level requirement, contractual criterion, or preregistered protocol can supply τ_j; an observed percentile cannot become its own pass value. The Sistema Informativo Territoriale (SI.Ter) procurement specification supplies a bounded front-end reference for interactions that do not depend on third-party systems[18]. That reference does not automatically transfer to mediated RVE workflows because their dependencies and observation boundary differ.

Repeated scenarios reduce but do not remove temporal confounding. Paired seeds align the synthetic choices, controlled ordering limits a fixed sequence effect, and run timestamps retain the chronology needed to detect environmental change. Remote observations within one run remain dependent. The final judgement therefore considers run-level dispersion and consistency of direction instead of relying only on a pooled request distribution.

Negative evidence remains part of the decision. A malformed event, missing terminal state, failed token transition, timeout, or generator overload can make a performance criterion not assessable while still exposing a framework or contract problem. Data-quality and semantic gates apply before aggregation, and their failures remain visible in the reported denominator.

2.5 Generic versus Domain-Specific Profiling

2.5.1 Modelling Distance and Semantic Fidelity

Apache JMeter and the Request Dataset Generator (RDG) represent different modelling paradigms rather than different protocol capabilities. JMeter pro-

[18] Azienda Zero, *Realizzazione di un Sistema Informativo Territoriale per gli Enti Sanitari della Regione Veneto*, pp. 56–57.

vides a mature request-centric model through samplers, controllers, extractors, variables, assertions, scripts, and distributed execution[19]. RDG uses a typed healthcare flow that integrates security, data, dependencies, scheduling, and evidence semantics.

Both paradigms issue concurrent SOAP/XML and REST/JSON requests. The decisive criterion is *modelling distance*: the additional interpretation and state logic required to express a documented regional workflow. A generic plan coordinates variables, extractors, assertions, controllers, and scripts to propagate security material, interpret patient-query outcomes, preserve context and document references, and block invalid transitions. RDG encodes these relations in the workflow vocabulary. The generic paradigm remains capable, but semantic fidelity depends on author-maintained conventions distributed across plan elements. This distribution increases the risk of cross-user state leakage, execution after transport-only success, or exclusion of failed prerequisites from the useful-flow denominator. Encoding these relations as workflow invariants reduces those failure points.

RDG also couples workload and evidence construction. Domain metadata creates synthetic patient and document contexts; stateful orchestration retains occurrence-local outputs and branches; asynchronous execution preserves concurrency among independent flows; and structured events distinguish technical attempts from terminal domain outcomes. The resulting model translates documented healthcare dependencies into executable, observable experimental units. Table 2.7 summarises the resulting trade-off.

[19] Apache Software Foundation. *Apache JMeter User's Manual: Elements of a Test Plan.* URL: https://jmeter.apache.org/usermanual/test_plan.html (visited on 08/04/2026); Apache Software Foundation. *Apache JMeter User's Manual: Component Reference.* URL: https://jmeter.apache.org/usermanual/component_reference.html (visited on 08/04/2026); Apache Software Foundation. *Apache JMeter User's Manual: Functions and Variables.* URL: https://jmeter.apache.org/usermanual/functions.html (visited on 08/04/2026); Apache Software Foundation. *Apache JMeter User's Manual: Remote (Distributed) Testing.* URL: https://jmeter.apache.org/usermanual/remote-test.html (visited on 08/04/2026).

Table 2.7: Generic and domain-specific profiling paradigms comparison. Source: Author's synthesis of the Apache JMeter User's Manual (Apache Software Foundation, accessed 2026) and the Anagrafe Zero v2.6, regional security v2.14, context-call v1.3, and Scryba Sign v16 specifications.

Dimension	**Generic request-centric paradigm**	**Domain-specific workflow paradigm**
Unit of analysis	Request or sampler sequence extended with user-defined state.	Typed information-dependent healthcare flow.
State semantics	Scope and lifetime emerge from variables, extractors, controllers, and custom logic.	Occurrence-local state and prerequisite blocking form explicit invariants.
Data semantics	Payloads remain generic unless the plan incorporates domain-specific generation and validation.	Synthetic patient and document metadata preserve the represented profile without using personal data.
Outcome semantics	Transport assertions require additional domain rules for application and flow completion.	Transport, application, and terminal-flow outcomes remain separate by construction.
Observability	Endpoint timing is native; workflow correlation and path differentials require explicit composition.	Correlated external events directly support surrogate path observation.
Evolution cost	Broad ecosystem and mature distributed facilities; domain contracts remain test-plan responsibilities.	Lower modelling distance for represented contracts; domain model requires maintenance when specifications change.

The comparison therefore concerns semantic risk and experimental traceability. A generic plan is attractive when the target is a stateless endpoint or a protocol-level baseline. A domain-specific workflow is attractive when the experimental outcome depends on ordered state transitions, domain metadata, application-level validation, and flow-level denominators. The Veneto use case belongs

to the second class because security, identity, context, document, and signing transitions determine whether useful work completes.

2.5.2 Justification of the Request Dataset Generator framework

RDG is methodologically justified when five conditions hold: the target process contains dependencies whose violation changes experimental meaning; synthetic data preserve contract structure without protected material; the research question concerns complete-flow behaviour and path differentials; and one evidence model connects offered load, state transitions, external timing, and terminal outcomes. A throughput comparison with JMeter requires matched workloads, equal host conditions, and a dedicated generator benchmark. A stateless generic baseline remains useful for calibration and protocol isolation, but it does not replace a session-aware workload for protected regional workflows. The choice relocates rather than eliminates complexity. A domain-specific model requires versioned transaction semantics, outcome definitions, scheduling controls, and event provenance. The same elements otherwise reappear as dispersed custom logic in a generic plan; RDG makes them inspectable at the abstraction level of the research question.

Scientific Rationale for Domain-Specific Profiling.

RDG responds to latency-aware distributed systems by combining quantile-based stochastic interpretation, full-stack FCT, matched Path-level Overhead, application-representative communication patterns, metadata-driven FHIR R4 generation, and session-aware orchestration. The absence of authorised traces makes controlled synthetic workflows the admissible internally valid input, while the regional security perimeter imposes Advanced Black-box with Surrogate Observability. RDG is therefore not merely custom software: it translates international methodological consensus into Regione Veneto's operational, semantic, and security constraints.

Summary of the Solution Space.

Under the available evidence perimeter, controlled Synthetic Workflows constitute this thesis's only experimentally valid instrument. Operational observation cannot isolate offered load from environmental variation; protocol benchmarks omit domain-critical dependencies; and trace replay requires authorised traces with reconstructable state and ordering. Since the permitted corpus contains no such trace, synthetic execution provides reproducible inputs while leaving external representativeness unproven. It also makes flow composition, arrival timing, randomisation, payload classes, and outcome rules explicit experimental factors.

The regional authorisation perimeter determines the observation method. Advanced Black-box with Surrogate Observability treats the protected path as a black box and correlates planned and actual dispatch times with client-boundary latency, application outcomes, dependency propagation, and flow completion. Scheduling drift, concurrency, and direct-versus-mediated differentials provide path-level evidence without assigning delay to inaccessible security, mediation, routing, or downstream phases. Mock events validate the measurement chain; claims about the live environment require authorised observations and complete provenance.

Semantic fidelity links these methodological choices. Fast Healthcare Interoperability Resources Release 4 (FHIR R4) payloads preserve the profile-relevant meaning of synthetic patient operations, while stateful orchestration keeps assertions, tokens, identifiers, document references, and signing state local to each flow occurrence. Sequential prerequisites remain ordered within workflows, while independent occurrences execute concurrently. Performance populations therefore represent completed or interrupted healthcare operations, not unrelated HTTP exchanges. Together, synthetic control, surrogate observability, and semantic fidelity define credible profiling requirements. Chapter 3 details the Request Dataset Generator framework's internal design and implementation, showing how these principles become an executable profiling instrument.

Chapter 3

Selected Approach

Chapter 1 defines the need for a controlled and traceable way of exercising healthcare-interoperability workflows. Chapter 2 then establishes the methodological requirements that follow from this need: the workload must preserve semantic dependencies, distinguish request- and flow-level observations, expose its temporal model, and keep client-side evidence separate from optional middleware-side measurements. This chapter presents the selected approach from the inside and records the evidence that is currently available.

The implementation described here is the Request Dataset Generator (RDG) repository at the revision inspected for this migration. The chapter distinguishes three claims that are easy to conflate. A feature is *implemented* when it is present in the current code; it is *functionally exercised* when a test or archived run covers it; and it is *experimentally evaluated* only when an identified configuration, execution log, normalised dataset, and metric report support the corresponding analysis. In particular, mock execution validates the framework pipeline but does not measure the capacity of a live regional or industrial service.

3.1 Technical Objectives and System Architecture

3.1.1 Architectural Objectives and End-to-End Pipeline

Technical objectives.

The framework translates the methodological requirements of Section 2.2.2 into five technical objectives. First, it must generate synthetic healthcare data without requiring personal data. Second, it must compose those data into ordered multi-step flows whose dependencies remain explicit. Third, it must schedule independent flow occurrences according to a configurable temporal model while preserving the order of steps within each occurrence. Fourth, it

must emit structured evidence at request, flow, and time-window granularity. Finally, it must make configuration, random seeds, execution mode, and output provenance recoverable from each run.

These objectives delimit the artefact. RDG is not a conformance-certification tool, a production audit system, or a clinical simulator. It creates and executes controlled workloads at the client boundary and can enrich the resulting records when a compatible middleware log is supplied. The absence of that second source remains visible in the normalised dataset rather than being replaced by estimated values.

End-to-end pipeline.

The command dispatcher in `main.py` exposes independent commands for patient generation, flow-dataset construction, timeline production, execution, log collection, and metric calculation. The `pipeline` command connects the same stages in memory and on disk:

$$\begin{aligned}&\text{configuration} \rightarrow \text{synthetic data} \rightarrow \text{timeline}\\ &\qquad\qquad \rightarrow \text{execution log} \rightarrow \text{normalised dataset} \rightarrow \text{metrics and reports.}\end{aligned}$$

This separation allows an intermediate artefact to be inspected or replayed without changing the responsibilities of the surrounding stages.

Figure 3.1: The Request Dataset Generator pipeline preserves a traceable chain from experiment configuration and Fast Healthcare Interoperability Resources Release 4 workload generation to asynchronous execution, JavaScript Object Notation Lines (JSONL) evidence, and statistical analysis. Source: Author's elaboration based on the current RDG source code.

Stage	Primary implementation	Produced or transformed evidence
Data generation	`fhir_patient_generator.py`	Synthetic Patient resources and transaction Bundles.
Workload composition	`request_dataset_generator.py`, `flow_orchestrator.py`	Weighted population of flows, ordered steps, identifiers, dependencies, and payload metadata.
Temporal scheduling	`timeline_generator.py`	Timestamped flow occurrences with base/burst provenance and experiment identifiers.
Execution	`traffic_exec_engine.py`	Mock or live step outcomes and JSONL request/response/flow events.
Collection	`log_collector.py`	Request, flow, and bucketed time-series populations; optional middleware enrichment.
Analysis	`metric_engine.py`	Machine-readable metrics, human-readable report, comma-separated-value tables, warnings, and optional graphs.

Table 3.1: Each pipeline stage owns one transformation and emits an inspectable artefact, which prevents generation, execution, and analysis concerns from being conflated. Source: Author's elaboration based on the current RDG source code.

3.1.2 Module Boundaries and Secure Transport

Repository organisation and dependency direction.

The top-level modules implement the pipeline stages, while the transaction package contains the transaction builders and the flow orchestrator. A transaction builder returns one executable step with its method, target selected from configuration, headers, body, transport profile, dependency specification, output-extraction rules, and—in mock mode—a synthetic response. The orchestrator adds correlation metadata and combines these steps into flow

types. The execution engine consumes this common shape; it does not contain healthcare-specific request templates.

This direction of dependency is important. Domain-specific contracts remain in the transaction modules, composition remains in the orchestrator, and temporal or analytical mechanisms operate on shared metadata. Adding a new transaction therefore does not require the scheduler or metric engine to understand its payload, provided that the module satisfies the common step contract and declares the outputs required by downstream steps.

Configuration and run provenance.

Configuration is loaded from JavaScript Object Notation (JSON) and merged with command defaults. Before a pipeline execution, the dispatcher establishes a scenario identifier, run identifier, execution identifier, and seeds for workload, timeline, and fault injection. It also resolves output paths under a run-specific directory and writes the effective configuration as `config.resolved.json`. The run directory can then contain the generated dataset, timeline, JSONL execution log, normalised metrics, metric reports, tabular summaries, and graphs. The effective configuration is more authoritative than the current scenario template when interpreting an archived run: templates may evolve after an execution, whereas the resolved copy records the parameters attached to that run. Credentials, security tokens, certificate material, private endpoints, and generated clinical-like values are deliberately excluded from this chapter. Reproducibility concerns non-sensitive parameters and provenance, not publication of protected configuration.

Mutual Transport Layer Security context management.

`MtlsContextFactory` separates cryptographic material from executable steps. A transaction declares only a named mutual-Transport Layer Security (mTLS) profile in its transport metadata; the runtime configuration retains certificate paths, verification policy, and any environment-variable reference used to re-

solve a certificate password. On first use, the factory constructs a Python `SSLContext`, loads either a Privacy-Enhanced Mail certificate chain or a Public-Key Cryptography Standards #12 container, and caches the context by profile name. Subsequent requests reuse the same context through the run-wide `aiohttp.ClientSession`, allowing eligible secure connections to benefit from session-level connection pooling without copying keys or passwords into generated datasets.

The factory returns no context when a transport profile is absent or disabled and raises an explicit error when an enabled profile lacks its certificate. This behaviour makes transport policy observable at configuration level while keeping sensitive material outside request logs and thesis artefacts. It also preserves the experimental distinction between protocol semantics and secure channel establishment: a failed mTLS setup is a transport outcome, not an application response from the regional gateway.

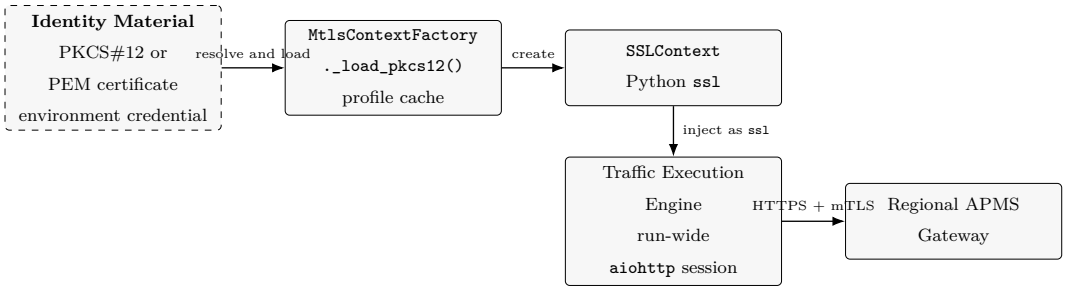


Figure 3.2: Secure Transport and mTLS Management Layer. The secure-transport layer keeps certificate material and password resolution outside generated workload steps: `MtlsContextFactory` creates and caches a profile-specific `SSLContext`, which the traffic engine supplies to its run-wide `aiohttp` session before protected regional requests. This separation makes channel-establishment failures distinguishable from application responses. Source: Author's elaboration based on the RDG implementation of `MtlsContextFactory` and `send_request()`.

3.2 Synthetic Inputs and Executable Transactions

3.2.1 Synthetic Resources and Transaction Heterogeneity

Synthetic Patient resources.

The data generator creates a FHIR Patient-shaped resource with synthetic identifiers, demographics, contact information, addresses, regional extensions, a contained practitioner, and a contained assistance contract. It can also construct the transaction Bundle used by the RVE-55 path and serialise generated structures as JSON or Extensible Markup Language (XML). Random generation removes the need for real patient records, while an optional identifier and a declared seed support controlled regeneration.

The generated structure is intended to exercise the contracts implemented by the framework. Its richness does not prove conformance with every cardinality, terminology rule, or business constraint of the applicable regional profile. That distinction mirrors the boundary established in Section 2.3.1: the code demonstrates what the workload can produce, while the primary specification remains authoritative for conformance.

Workload population and correlation model.

`build_workload` accepts the number of source flows, a weighted `flow_mix`, an execution mode, the complete configuration, and an optional seed. Weights are normalised and sampled to choose a flow type; a new synthetic patient is generated for each source flow. The resulting flow carries a flow identifier, a type, and an ordered list of steps. Every step has its own request identifier and index, while the timeline later adds a distinct `flow_occurrence_id` whenever a source flow is scheduled. This last identifier prevents repeated uses of the same source flow from being merged during log reconstruction.

The current implementation defines the seven flow types in Table 3.2. The table reports software coverage, not equal validation status. In particular, a zero default weight or presence in the map does not establish that a path has

been exercised against a live provider.

Flow type	Ordered transaction modules	Steps
`FLOW_CONTEXT_CALL`	RVE-1.b → RVE-121 → RVE-130	3
`FLOW_PATIENT_QUERY`	RVE-1.b → RVE-54	2
`FLOW_PATIENT_CREATE`	RVE-1.b → RVE-55	2
`FLOW_ITI_18`	RVE-1.b → ITI-18	2
`FLOW_ITI_43`	RVE-1.b → ITI-43	2
`FLOW_PATIENT_QUERY_MIDDLEWARE`	RVE token → gateway RVE-54 → gateway RVE-55 → gateway RVE-57 → gateway RVE-100	5
`FLOW_SCRYBASIGN_SIGN`	GetUserInfo → SignOneDoc	2

Table 3.2: The orchestrator maps heterogeneous regional, Integrating the Healthcare Enterprise, mediated, and digital-signature transactions to one ordered step contract; step count remains part of every comparison. Source: Author's elaboration based on the current RDG source code, the regional specifications cited in Chapter 2, and the Scryba Sign developer guide.

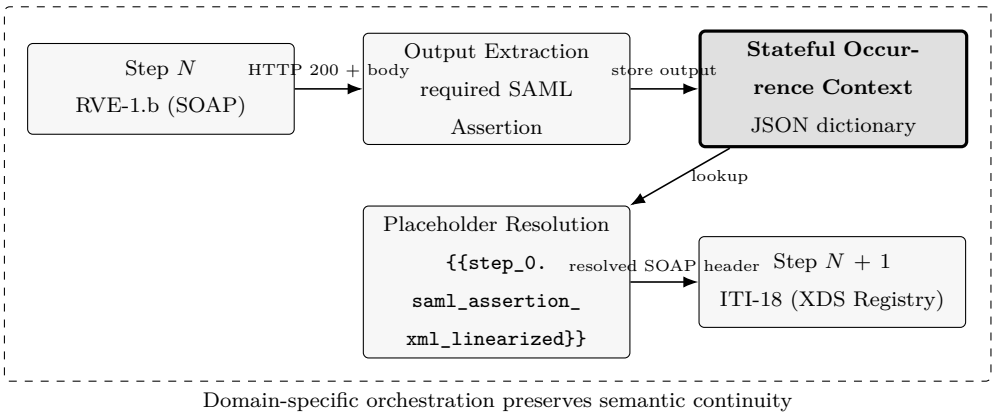


Figure 3.3: Semantic Transformation and Placeholder Resolution. The RVE-1.b response becomes an executable dependency rather than a loose ordering convention: the engine extracts the required SAML Assertion into the occurrence-local context, resolves the exact ITI-18 placeholder, and prevents transmission while a required value remains unresolved. Workflow success therefore depends on semantic continuity across the two protocol steps. Source: Author's elaboration based on the current RDG implementations of `rve_1b.py`, `iti_18.py`, and `traffic_exec_engine.py`.

RVE, ITI, and mediated paths.

The direct regional paths combine an initial assertion-producing transaction with context, patient-registry, or document transactions. The two ITI flows use

dedicated transaction modules after the initial integration step. The mediated patient path has a different structure: it obtains a middleware token and then executes the configured gateway scenarios for RVE-54, RVE-55, RVE-57, and RVE-100. This five-step path is therefore not a request-for-request replacement for the two-step direct patient query. Comparisons between their completion times must acknowledge the different step count and contracts.
The code confirms that RVE-57 is present in the mediated flow, whereas the permitted source corpus contains no authoritative specification that establishes its complete domain semantics. The chapter therefore treats RVE-57 as an implemented transaction label and excludes it from claims of semantic equivalence with the direct path described in Section 2.3.

Digital-signature provider flow.

`FLOW_SCRYBASIGN_SIGN` composes two Simple Object Access Protocol (SOAP) POST steps in a fixed order: the provider user-information operation and the single-document signing operation. The shared `scrybasign_common.py` module centralises endpoint selection, mTLS-profile resolution, and construction of the Hypertext Transfer Protocol (HTTP) `Authorization` header. In particular, `resolve_auth_header()` reads a scoped username and password, optionally resolves the password from an environment variable, encodes the credential pair in Base64, and returns the Basic-authentication value consumed by both builders. The generated dataset therefore contains the application header but not the certificate material managed by `MtlsContextFactory`.
The signing envelope carries its document in the Base64 `docBin` field. The current builder embeds a fixed test artefact rather than encoding a generated clinical document, and it declares no response dependency between `GetUserInfo` and `SignOneDoc`. The credential block also exposes an optional one-time password (OTP) element, but the builder leaves it empty: `resolve_auth_header()` implements transport-level Basic authentication and does not generate or verify an OTP. This distinction keeps strong signing authentication inside the

provider contract instead of misrepresenting it as an implemented framework capability.

The provider documentation distinguishes synchronous and asynchronous signing surfaces.[1] The current flow map instantiates only the synchronous `SignOneDoc` path. The workload model can represent the two interaction styles as distinct flow types, step sequences, and completion criteria, but an asynchronous signing experiment requires explicit enqueue, status, or notification transaction modules that the inspected source does not contain. Mock envelopes exercise the implemented two-step synchronous shape without contacting the provider, while the sanitised live corpus establishes client-boundary execution coverage without proving provider-side response semantics or authorisation. Section 3.6 retains that evidential boundary.

Step dependencies and the mock/live boundary.

A flow is more than a list of independent Hypertext Transfer Protocol (HTTP) requests. A step may declare that a field comes from an output of an earlier step. In live mode, downstream headers, bodies, and Uniform Resource Locators (URLs) can contain scoped placeholders; the execution engine resolves them from the occurrence-local context only after processing the preceding response. Required outputs are validated before the chain continues. An unresolved placeholder or missing required output is logged as a failure instead of sending an incomplete dependent request.

In mock mode, transaction builders attach synthetic responses and the orchestrator propagates their declared outputs while constructing the chain. Execution then consumes those responses without opening an HTTP session. This mode tests composition, scheduling, failure handling, logging, collection, and metric production under controlled conditions. It cannot measure network, gateway, provider, or production-system behaviour. Dry-run is a separate ex-

[1] Medas S.r.l., *Scryba Sign 3.x Developer's Guide*.

ecution branch that records a successful empty interaction without sending requests and is distinct from both mock-contract validation and live-service evidence.

3.3 Traffic Execution and Evidence Pipeline

3.3.1 Temporal Scheduling and Stateful Execution

Temporal model and occurrence generation.

The timeline generator separates flow content from arrival time. It supports a constant rate, a configurable step profile, and a synthetic time-banded clinical profile. For a variable intensity $\lambda(t)$, arrivals are produced through Lewis–Shedler thinning: candidates are sampled from an upper-bound homogeneous process and retained according to $\lambda(t)/\lambda_{\max}$. Optional bursts add a configurable number of closely spaced occurrences around selected baseline arrivals. Every event records whether it is a base or burst arrival and, where applicable, its burst identifier.

The generator samples from the already constructed source dataset according to its empirical flow-type distribution. It therefore changes the temporal population without modifying transaction definitions. A seed makes workload and timeline sampling repeatable for a fixed code and configuration, but it cannot make the latency of live remote services deterministic. The nominal duration and rate also do not fix the final number of arrivals, which remains a realisation of the stochastic process.

Scheduling and concurrency semantics.

The execution engine follows scheduled timestamps using an asynchronous task per flow occurrence. A semaphore limits concurrent flows and encloses the whole flow rather than a single request. Inside that boundary, steps are executed in order and share only their occurrence-local output context; separate flows can overlap. The engine stops a chain at the first non-success status or

processing error, records the failing step, and leaves the remaining downstream steps unexecuted.

Live execution opens an HTTP client session, resolves dependencies, optionally uses the configured proxy and mTLS context, sends the request, extracts declared outputs, and validates required fields. Mock execution reads the synthetic response attached to each step. A deterministic fault injector can apply declared error, timeout, or latency-spike rules by transaction, flow type, or step. These injected outcomes evaluate framework behaviour under controlled faults; they are not observations of provider reliability.

3.3.2 Configuration Interface and Evidence Processing

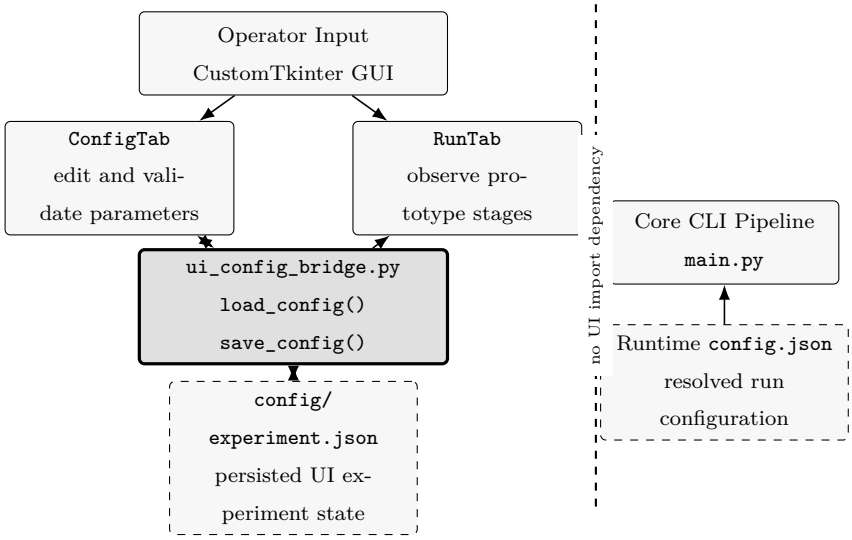


Figure 3.4: Experiment Persistence and UI Bridge Architecture. The desktop interface reads and writes its experiment state only through `ui_config_bridge.py`: `ConfigTab` changes persisted parameters, while `RunTab` reloads them to drive its stage-level controls and status view. The core command-line pipeline remains outside this dependency boundary and resolves its own runtime configuration, so interface persistence does not couple the executable engine to CustomTkinter. Source: Author's elaboration based on execution of the current RDG interface and inspection of `ui_config_bridge.py`, `config_tab.py`, `run_tab.py`, and `main.py`.

User-interface configuration bridge.

The optional desktop interface separates experiment persistence from visual state through `ui_config_bridge.py`. Its `load_config()` function merges the JSON file with explicit defaults and tolerates a missing or invalid file; `save_config()` serialises the values selected by the operator. The dependency remains one-way: interface modules import the bridge, whereas the command-line pipeline does not depend on the user-interface package. The bridge consequently provides a stable persistence boundary rather than a second configuration model.

The three tabs expose complementary operational views. `ConfigTab` collects experiment name, duration, arrival intensity, synthetic-patient count, active transactions, mediated-flow type, and target environment, then persists them through the bridge. `RunTab` presents stage-level controls, thread-safe progress, and a live status log. In the inspected revision its workers simulate the four stage transitions and write demonstration outputs; they do not invoke the verified `main.py` pipeline. `DashboardTab` reloads metric and report files after a run and presents throughput, P95 latency, error rate, request count, per-flow distributions, and the Markdown report. The interface is therefore a functional prototype for configuration and immediate inspection, while reproducible execution remains attached to the command-line pipeline and its resolved configuration.

Structured execution events.

The execution logger emits three JSONL event types. A `request_sent` event records experiment identifiers, flow occurrence, request and step identifiers, transaction, payload size, scheduled and actual start times, queue delay, active flows and steps, and burst provenance. A `response_received` event records status, response size, client latency, success, error, and fault-injection provenance. A `flow_completed` event records total duration, success, first failed

step, declared step count, and burst provenance.

This schema implements the distinction between an interaction and an application flow established in Chapter 2. It also records enough provenance to stratify observations by scenario, flow, transaction, arrival type, or execution. Payload contents, assertions, tokens, credentials, and certificate data are not required by the analytical schema and must not be exported as thesis evidence.

Normalisation and surrogate enrichment.

The collector parses the framework JSONL stream, pairs sent and received events by occurrence, reconstructs completed flows, and aggregates a bucketed time series. Its output contains four top-level parts: metadata, requests, flows, and timeseries. Request records retain client latency, outcome, concurrency, scheduling, size, and provenance. Flow records retain completion time, executed steps, skipped downstream steps, and outcome. Metadata records the source files and the observed time range.

A second parser can ingest a compatible middleware JSONL stream. Matching records may add gateway and upstream latencies, retry count, circuit-breaker state, and infrastructure observations. When no middleware record is available, these fields remain null. The present archived runs report zero middleware events, so they support the client-boundary layer of Advanced Black-box with Surrogate Observability. No gateway/upstream decomposition or infrastructure correlation derives from those records.

Metric and reporting engine.

`MetricsCalculator` transforms the normalised request, flow, and time-bucket populations into eighteen measures grouped into performance, reliability, resilience, scalability, and observability. Performance separates Flow Completion Time (FCT) from step latency and reports request and flow throughput, burst strata, and Path-level Overhead (ΔL) for compatible direct and mediated populations. Reliability retains error, timeout, retry, and flow-success

rates; resilience derives recovery episodes, propagation, and bucket-level availability; scalability relates available resource and concurrency observations to load; observability measures event correlation and scheduling fidelity.

`StatisticsHelper` sorts each eligible population and calculates its size, mean, median, standard deviation, range, and P50, P75, P90, P95, and P99 through linear interpolation between adjacent ranks. Median, P95, and P99 retain the centre and tail of the empirical distribution, in line with the latency-analysis rationale established by Iorio et al. and Akram et al.[2] The `WarningDetector` then evaluates the calculated structure against configurable thresholds for tail latency, error rate, availability, resource saturation, and scheduling drift; it also flags a P99/P50 ratio above the configured skew rule. JSON and Markdown reports, comma-separated-value tables, and optional graphs preserve both the statistic and its population. These warnings automate inspection but remain software signals, not clinical, contractual, or production acceptance criteria.

3.4 Evaluation Protocol and Experimental Design

3.4.1 Evaluation Questions and Construct Validity

Evaluation questions.

The implemented architecture and the evidence required by the problem statement lead to four evaluation questions:

1. Can RDG preserve ordered transaction dependencies and isolate the state of concurrent flow occurrences in both mock and live modes?

2. Can a declared flow mix and temporal model be reproduced with traceable identifiers, seeds, configuration, and artefacts?

3. Can execution events be transformed into consistent request-, flow-, and time-window populations without filling unavailable middleware fields?

[2]Iorio, Risso, and Casetti, "When Latency Matters: Measurements and Lessons Learned"; Akram, Akram, and Rehman, "Numerical Evaluation of Network Latency and Throughput in Distributed Systems".

4. Within an approved scenario, how do latency distributions, throughput, reliability, queueing, and flow completion change with flow composition, offered load, bursts, or injected faults?

The first three questions concern functional and methodological validity. The fourth concerns experimental behaviour and must be answered separately for each execution mode and observed system boundary.

Critical evaluation of the architecture.

The pipeline provides strong separation between domain contracts, temporal load, execution, and analysis. It also makes intermediate artefacts inspectable and allows the same collector and metric engine to process mock and live logs. The main architectural trade-off is that a generic step contract can preserve ordering and metadata without establishing semantic equivalence among flows. A two-step direct patient query and a five-step mediated path can both be executed, but their difference cannot be attributed solely to middleware.
The execution semaphore applies to complete flows. This protects within-occurrence dependencies and makes the number of active flows explicit, but it also means that long or blocked flows retain a concurrency slot. This is part of the evaluated execution model and does not represent an unqualified model of provider-side concurrency. Similarly, the client-side clock measures scheduling and request completion at the generator; internal service queues remain outside that boundary unless a verified second source is available.

Advanced Black-box with Surrogate Observability.

Framework evidence always records scheduled and actual dispatch, client-side step latency, response outcome, flow completion, and client-observed concurrency. Optional surrogate evidence records middleware intervals, retries, circuit-breaker state, or infrastructure measurements. A join is valid only when identifiers, occurrence semantics, clocks, and measurement boundaries are compatible; join coverage is a data-quality measure rather than an assumption.

Unsynchronised series are not aligned visually or subtracted across different clock and boundary semantics. A statistical association supports a diagnostic hypothesis, whereas causal attribution requires controlled variation and compatible measurements. The available run corpus contains framework evidence only, so Advanced Black-box with Surrogate Observability preserves the internal path as an explicit attribution boundary.

3.4.2 Variables and Reproducibility Controls

Variables and controlled factors.

The independent variables exposed by the framework include execution mode, flow mix, source-dataset size, temporal profile and intensity, burst parameters, concurrency limit, timeout, and declared fault-injection rules. Dependent variables include step and flow latency distributions, request and flow throughput, error and timeout rates, flow-success rate, scheduling drift, queue delay, active flows and steps, recovery episodes, and evidence completeness.

At minimum, a comparison must hold constant the code revision, transaction modules, execution mode, credentials and transport policy where applicable, flow mix unless it is the factor under study, seed policy, duration, concurrency limit, metric implementation, and observation boundary. Payload-size and step-count distributions must be reported when they differ across compared flows. Live tests additionally require a documented hardware, operating-system, network, remote-environment, and time-of-execution context.

Reproducibility protocol and campaign controls.

An analysable run must retain the code revision, resolved configuration, dataset, timeline, execution log, normalised dataset, metric report, and any excluded or failed-run record. Scenario, run, and execution identifiers must agree across these artefacts. Seeds must be recorded separately for workload, timeline, and fault injection. Results must identify the execution mode and must never combine mock and live samples in one distribution.

The current archive satisfies much of the artefact chain for individual runs, but it does not constitute the final approved experiment campaign. The sanitised batch specifies 30 repetitions for each of ten scenarios, whereas the available complete population ranges from five to fifteen runs per scenario and totals 74 runs. Provenance identifies source commit `f08fb53fd4ad11ef803642c0a066e9a6d9e2b327` as the RDG checkout observed at extraction time, rather than as a revision embedded in the batch manifest.

Construct validity remains independent of whether the incomplete campaign supports a final performance comparison. The framework already exposes the controls required to define workload, scheduling, concurrency, fault policy, and evidence provenance. Performance inference additionally depends on an approved warm-up and cool-down policy, minimum populations, hardware and operating-system inventory, network-path and clock-synchronisation evidence, remote-environment version, run-exclusion rules, and an immutable revision in the campaign manifest. Since the available archive does not provide this full set, the chapter treats those items as limits of the observed campaign rather than gaps in the implemented profiling capability.

3.5 Executed Scenarios, Performance Indicators, and Quality Controls

3.5.1 Scenario Evidence and Operational Populations

Declared scenarios versus executed evidence.

The repository contains configuration templates for constant baselines, step-load ramps, bursts, Path-level Overhead (ΔL), fault injection, flow-type comparisons, a small live smoke test, and an extreme stress profile. Their presence describes intended experiment capabilities. Only a scenario with a run directory and a coherent artefact chain is treated here as executed. The resolved configuration in each run directory determines its mode and parameters.

Table 3.3 inventories the six available run directories. Five are explicitly mock

executions. The sixth is recorded as a live, three-flow smoke run, but its size and coverage make it suitable only as limited functional evidence.

Archived scenario	Mode	Flows	Requests	Observed flow types
`baseline_mock`	mock	12	26	context, patient query, patient create
`burst_mock`	mock	136	321	context, patient query, patient create
`direct_vs_middleware_mock`	mock	76	287	direct patient query, mediated patient path
`error_injection_mock`	mock	65	151	context, patient query, patient create
`flow_type_comparison_middleware`	mock	45	225	mediated patient path only
`live_smoke_small`	live	3	6	direct patient query only

Table 3.3: The migration archive links each executed scenario to a resolved configuration and normalised dataset, thereby supporting functional traceability without implying a balanced final campaign. Source: Author's elaboration based on the archived RDG run directories and their resolved configurations.

The six-run migration archive does not cover ITI-18, ITI-43, the `FLOW_SCRYBASIGN_SIGN` flow, a step ramp, or the extreme-stress profile. The subsequent sanitised live corpus extends transaction coverage and contains five complete `load_ramp` runs, five `stress_extreme` runs, six `flow_type_comparison_direct` runs, fifteen `flow_type_comparison_middleware` runs, and live outcomes for ITI-18, ITI-43, and both Scryba Sign transactions. The two inventories have different roles: Table 3.3 documents the minimal artefact chain used for functional migration, whereas the sanitised corpus documents broader live execution.

The campaign manifest defines 300 runs across ten scenarios; the available corpus contains 145 started runs and 74 complete runs. This imbalance preserves observed outcomes but prevents the incomplete sample from acting as a final, balanced comparison. It is a limitation of the experimental campaign, not of the framework's ability to generate, execute, and normalise the declared scenario families.

Operational performance populations.

Flow completion time is calculated over completed flow records and must be stratified by flow type, outcome, and mode. Step latency is calculated over paired request/response occurrences and is stratified at least by transaction and flow type. Request throughput counts completed requests per observed second; flow throughput counts completed flow occurrences over the same run interval. The two are reported together because flow types have different step counts.

Reliability uses request errors, timeouts, and completed-flow outcomes. Failure at a step is associated with skipped downstream steps rather than fabricated requests. Scheduling fidelity uses the difference between planned and actual client dispatch together with queue delay and active-flow/step counts. Burst analysis uses the explicit base/burst labels. Middleware breakdown, resource saturation, and retry or circuit-breaker measures are valid only for records actually enriched by a second source.

3.5.2 Quality Controls and Acceptance Boundary

Data-quality controls.

Before interpretation, each run is checked for:

- an available resolved configuration and explicit execution mode;
- consistent scenario, run, execution, flow-occurrence, flow, request, and step identifiers;
- paired sent/received events and an explicit flow-completion event;
- recorded population sizes, time range, units, and flow-type coverage;
- explicit fault-injection and burst provenance;
- separate mock and live populations;
- the count and coverage of middleware-enriched records;

- retained failed or incomplete observations rather than silent imputation.

All six normalised migration datasets identify their framework log, and their request records are associated with reconstructed flows. They also report zero middleware events and zero requests with middleware timing fields. This is a valid client-boundary condition for Advanced Black-box with Surrogate Observability, but it prevents cross-layer decomposition. A surrogate analysis therefore remains conditional on a verified middleware log with compatible occurrence keys, documented measurement boundaries, clock evidence, and quantified join coverage.

Acceptance criteria.

The archived reports contain configurable warning values and the engine produces warnings when those values are crossed. They are useful for automated inspection but are not clinical, contractual, or production-service acceptance limits. The experiment can currently judge internal consistency, functional completion, reproducibility of mock inputs, and descriptive behaviour within a scenario. It cannot judge production suitability against an approved service level.

The chapter consequently applies acceptance at construct level: identifiers remain coherent, dependencies resolve within an occurrence, mock and live populations remain separate, failed observations remain explicit, and every reported statistic identifies its source population. Production acceptance remains outside the current evidential boundary because the campaign does not provide domain-supported thresholds, required uncertainty, minimum sample sizes, a complete repetition count, or an ex ante scenario–indicator–population–threshold–evidence matrix. These controls qualify the strength of experimental conclusions rather than the existence of the implemented measurement pipeline.

3.6 Evidence, Discussion, and Design Guidelines

3.6.1 Functional and Quantitative Evidence

Functional evidence.

The recorded functional-validation run exercises placeholder resolution, supported flow composition, Integrating the Healthcare Enterprise profiles, Scryba Sign request construction and ordering, timeline occurrence identity, burst tagging, configuration validity, and pipeline command dispatch. It contains 29 tests: 28 complete successfully, while the graph/table-output test stops because the optional plotting dependency is unavailable. The successful population validates the implemented contracts covered by those tests and the transformation from generated flows to normalised evidence. The remaining environmental error does not invalidate metric calculation, but it prevents the record from constituting an all-green release criterion. The current source contains a larger test inventory, so the 28-test result remains evidence for the recorded migration snapshot rather than an assertion about every later checkout.

The archived mock runs further validate that generated flows can traverse the timeline, executor, logger, collector, and metric engine. The small live run records three successful direct patient-query flows, six successful requests, and the expected two-step transaction shape. Its population is too small and its environmental context too incomplete for performance inference; it is reported only as limited live-path functional evidence.

Mock pipeline results.

Table 3.4 reports selected outcomes from the five mock runs. Values describe execution of synthetic responses and framework-side processing on the recorded host. They must not be read as regional-service, gateway, middleware, or provider latency.

Scenario	FCT P50 ms	FCT P99 ms	Requests/s	Error %	Flow success %
Baseline mock	0.462	0.859	11.770	0.000	100.000
Burst mock	0.694	2.134	27.821	0.000	100.000
Direct/mediated mock	1.796	3.033	18.162	0.000	100.000
Fault-injection mock	0.839	50.930	12.768	5.960	86.154
Mediated-flow mock	2.259	6.599	24.534	0.000	100.000

Table 3.4: Selected mock-run outcomes demonstrate framework-side scheduling, failure retention, and metric generation; they do not estimate regional-service latency. Flow Completion Time (FCT) is measured over completed occurrence records. Source: Author's elaboration based on archived RDG mock reports generated by `metric_engine.py`.

The baseline and burst runs show that burst-labelled occurrences are processed and remain available for stratified analysis. The increase in reported P99 in the burst run is an observation of this mock execution, not evidence of remote system degradation. Only one seed and one execution are available, and the two runs do not establish a distribution of run-to-run variability.

The mock Path-level Overhead (ΔL) report contains 31 direct and 45 mediated flow occurrences. Their mean completion times are respectively 0.781 ms and 2.166 ms, with P95 values of 1.147 ms and 2.961 ms. The report calls the difference an *apparent overhead*; this wording is essential. The paths have two and five steps respectively, use synthetic responses, and have no middleware-internal observations. The difference validates stratified reporting but cannot isolate middleware cost.

The fault-injection run contains 65 flows and 151 executed requests. Nine flows fail after declared synthetic faults, giving a flow-success rate of 86.154% and a request error rate of 5.960%. The report identifies two recovery episodes with a mean duration of 3.5 s. These values demonstrate that the pipeline retains injected failures, interrupts downstream steps, and computes resilience summaries. They do not estimate the failure or recovery behaviour of a live service.

Surrogate-observability result boundary.

Although the runs used to calculate Path-level Overhead (ΔL) and the mediated-only runs exercise the five-step gateway path, their normalised datasets contain no middleware events. Consequently, gateway latency, upstream latency, retry count, circuit-breaker state, central processing unit (CPU), memory, and internal connection measurements are unavailable. The only defensible comparison is between client-observed flow shapes in mock mode. The surrogate layer of Advanced Black-box with Surrogate Observability remains unobserved. Cross-layer latency or resource relationships therefore require matched successful direct and mediated live populations, a documented middleware log, verified occurrence-level joins, compatible clocks, and explicit measurement boundaries; none belongs to the current result set.

3.6.2 Interpretation and Contribution Boundary

Relation to the state of the art.

The available evidence supports the methodological conclusions of Chapter 2 at the level of the artefact. Step and flow populations are both retained, percentiles expose distribution tails that a mean hides, and structured occurrence identifiers prevent repeated source flows from being collapsed. The evidence does not yet support the stronger empirical comparison with the reviewed literature: different systems, payloads, network paths, and experimental designs preclude transfer of their quantitative findings.

In particular, the mock tail and fault-injection observations illustrate why P95/P99 and failed-flow populations are useful, but do not validate a production scheduling or capacity rule. Likewise, the small live smoke run does not establish that lower-level optimisations improve end-to-end performance. Those questions require the missing repeated live campaign and declared acceptance framework.

Design and evaluation guidelines.

The implementation and current evidence support the following conservative guidelines:

1. preserve transaction, step, flow, and occurrence identities from generation through reporting;
2. treat step order and dependency propagation as correctness invariants, while applying concurrency between independent flows;
3. record resolved configuration and independent seeds with every run;
4. separate mock, dry-run, and live evidence in storage and analysis;
5. report request and flow throughput together because flow shapes have different step counts;
6. qualify Path-level Overhead (ΔL) by semantic equivalence, step count, payload, and observation boundary;
7. leave unavailable middleware and infrastructure fields null and quantify their coverage;
8. distinguish configurable warnings from approved acceptance thresholds;
9. calibrate flow mixes and arrival profiles before claiming representativeness of operational traffic;
10. derive production or capacity guidance only from repeated, controlled, appropriately authorised live experiments.

Residual limitations and contribution boundary.

The present chapter can substantiate the architecture, current transaction map, mock/live boundary, evidence pipeline, archived mock validation, and one limited live patient-query smoke run. It does not substantiate production capacity,

provider reliability, causal middleware attribution, or operational traffic representativeness. It establishes execution coverage, rather than a complete performance characterisation, for the Integrating the Healthcare Enterprise and Scryba Sign paths. These limits preserve the distinction between software capability and experimental evidence established at the beginning of the chapter. The newer sanitised live corpus provides repeated execution evidence for ITI-18 (21,722 outcomes, including 3,395 successes), ITI-43 (21,514 outcomes, including 21,371 successes), `SCRYBASIGN-GET-USER-INFO` (17,838 successes), and `SCRYBASIGN-SIGN-ONE-DOC` (17,838 outcomes, including 17,015 successes). The planned protocol remains incomplete, and the sanitised corpus contains no middleware-side surrogate evidence. Consequently, these counts support transaction-outcome coverage within the 74 complete runs but not final cross-layer, Path-level Overhead (ΔL), or capacity conclusions.

The contribution therefore closes at two levels. At artefact level, the chapter establishes a configurable, stateful, asynchronous, and observable profiling pipeline. At experimental level, it reports only hypotheses and populations supported by retained evidence, including failed runs and incomplete scenario coverage. Final production rules remain conditional on a balanced authorised campaign, ex ante acceptance criteria, uncertainty reporting, and compatible surrogate observations; the absence of those campaign controls does not reduce the implemented framework to an undocumented prototype.

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